TTS-HUB: LEVERAGING MODULAR LORAS AND ARITHMETIC COMPOSITION FOR CONTROLLABLE TEXT-TO-SPEECH

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Paper under double-blind review

ABSTRACT

Controllable text-to-speech (TTS) aims to generate speech from text while allowing control over prosodic and speaker-related attributes such as pitch, age, and accent. Existing controllable TTS methods primarily rely on natural language prompts to guide the synthesis process or utilize reference audio cloning to achieve control. However, prompt-based approaches often struggle with the cross-modal semantic gap between textual descriptions and intended speech attributes, leading to imprecise and coarse-grained control. Conversely, cloning methods depend heavily on reference audio samples and struggle to generalize beyond the characteristics seen in those samples, resulting in limited flexibility. To overcome these challenges, this paper proposes TTS-Hub, a novel controllable TTS framework that employs modular Low-Rank Adaptation (LoRA) components and their arithmetic-based composition to achieve fine-grained and flexible controllable TTS. Specifically, we construct a comprehensive Data Hub, which covers 6 high-level attribute categories and 32 fine-grained speech attributes. Leveraging this attribute-specific data, we fine-tune two mainstream TTS frameworks to obtain a corresponding LoRA Hub, where each modular LoRA is specialized for a specific speech attribute. At inference time, TTS-Hub selects the required LoRA modules and combines them through simple arithmetic composition to produce a fused LoRA that simultaneously encodes multiple attribute representations, enabling flexible and extensible multi-attribute control without retraining the backbone. Extensive experiments show that individual LoRAs provide precise single-attribute control, while arithmetic composition yields flexible and interpretable multi-attribute speech and consistently outperforms prompt-based baselines. Code and data are available in supplementary materials.

1 INTRODUCTION

Text-to-speech (TTS) aims to convert textual input into natural-sounding speech and has rapidly advanced from neural network-based architectures (e.g., Tacotron 2 (Shen et al., 2018), FastSpeech (Ren et al., 2019), NeuralSpeech (Tan et al., 2024), and CosyVoice (Du et al., 2024a)) to large language model (LLM)-based synthesis systems such as GPT-4o-mini-TTS (OpenAI, 2025), Gemini-2.5-TTS (Comanici et al., 2025), and Qwen-TTS (Qwen, 2025). To meet the growing demand for personalized synthesis, controllable TTS systems (Ji et al., 2025; Wang et al., 2025; Huang et al., 2024c;b) have become increasingly essential. These systems enable users to precisely specify speech attributes, including accent, age, and emotion, thus facilitating diverse applications such as customized audiobooks, personalized voice assistants, and virtual game character creation.

Current Controllable TTS approaches are dominated by two paradigms, each with significant limitations. Natural-language prompting (Guo et al., 2023; Yang et al., 2024; Ji et al., 2024a) guides speech synthesis through intuitive instructions (e.g., "read this like a sad teenager with a Scottish accent"), but the inherent cross-modal semantic gap requires the model to infer distinctive acoustic features from vague textual descriptions. These features are often only weakly expressed or partially realized in the generated speech, resulting in coarse and imprecise control over individual attributes (Xie et al., 2024). Reference-audio cloning (Hao et al., 2025; Ju et al., 2024; Jiang et al., 2024) alleviates this ambiguity by conditioning on a provided reference clip. However, it requires

a precisely matched reference for every desired attribute combination, rendering it impractical for diverse or composite requests such as "elderly, cheerful, Mandarin accent" (Barakat et al., 2024). Consequently, there is an urgent need for a fine-grained and flexible controllable TTS solution.

A promising solution emerges from Low-Rank Adaptation (LoRA) (Hu et al., 2022), a parameter-efficient fine-tuning technique that keeps the pretrained backbone frozen while inserting compact low-rank adapter matrices that are learned to adapt to specific tasks and capabilities. The modular architecture of LoRA has sparked further research on the synergistic composition of multiple trained LoRAs, showing that composing several pretrained LoRAs can improve cross-task generalization (Huang et al., 2024a; Wu et al., 2024). This composable modularity motivates us to investagate LoRAs and their composition for controllable TTS.

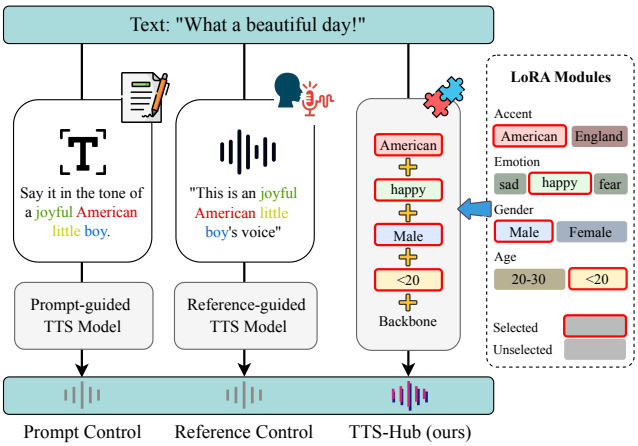


Figure 1: Overview of two mainstream Controllable TTS Paradigms and Our Proposed TTS-Hub Framework. Existing methods rely on natural-language prompts (coarse and imprecise due to cross-modal semantic gaps) or reference audio cloning (inflexible and hard to generalize). In contrast, TTS-Hub achieves fine-grained, flexible, and extensible control via composing attribute-specific LoRA modules.

To this end, we propose TTS-Hub, which treats each LoRA as an attribute specialist and achieves fine-grained and flexible control by selecting and composing the relevant LoRAs at inference. Specifically, we first collect four publicly available TTS datasets and reconstruct the samples into a comprehensive data hub, covering 6 high-level attribute groups (accent, age, emotion, gender, energy, pitch) and 32 fine-grained speech attributes. Leveraging these attribute-specific subsets, we build a LoRA hub by fine-tuning one adapter in a specific attribute. During inference, TTS-Hub selects the adapters that match a user's request, combines them into a single composite LoRA when multiple attributes are needed, and inserts this adapter into the frozen backbone to generate speech. Extensive experiments demonstrate that: (i) individual adapters can precisely control their target attributes; and (ii) arithmetic combination allows intuitive mixing, attenuation, or amplification of attributes, providing a flexible and interpretable approach to multi-attribute control. In addition, TTS-Hub is inherently extensible: new attribute adapters can be added to the hub, thereby continuously expanding the system's expressive capabilities. To the best of our knowledge, we are the first to explore LoRAs and their composition for controllable TTS.

Our main contributions are as follows.

- We present TTS-Hub, the first framework that formulates controllable TTS as arithmetic-based composition of LoRA adapters, enabling fine-grained and flexible control.
- We construct and release a comprehensive and easily-extensible hub of attribute-specialist adapters for two mainstream backbones, enabling reproducible research and practical deployment.
- Extensive experiments demonstrate that individual adapters deliver precise single-attribute control, whereas arithmetic composition yields interpretable multi-attribute synthesis.

2 Related Works

Controllable TTS.Recent advances in controllable text-to-speech (TTS) systems, such as VALL-E R (Han et al., 2024), CosyVoice (Du et al., 2024b), and GLM-4-Voice (Zeng et al., 2024), have made significant strides in generating high-fidelity speech with some degree of style control, driven by large-scale models and extensive training data. Systems like PromptTTS (Guo et al., 2023) use natural-language prompts for acoustic attribute control (e.g., pitch, timbre), while (Liu et al., 2024) supports accent control and (Bott et al., 2024) emotion control. However, challenges remain in achieving fine-grained controllability. Current models struggle to synthesize specific voices, such

as a 60-year-old Irish speaker, or combine nuanced attributes flexibly. Moreover, reference-audio cloning in systems like (Ruggiero et al., 2021) captures speaker traits but is impractical for complex attribute combinations, particularly in real-time scenarios. Consequently, most systems are limited to replicating example styles or offering only a narrow set of attributes.

LoRA-based Fine-tuning for TTS. Low-Rank Adaptation (LoRA) (Hu et al., 2022) provides a parameter-efficient method for fine-tuning large pretrained models by learning low-rank weight updates while freezing the backbone parameters. This approach has been widely used in speech synthesis for efficient feature learning. For example, (Qi et al., 2024) modeled emotional features with LoRA modules, (Bondaruk & Kubiak, 2025) enhanced multilingual expressiveness, and (Lou et al., 2024) improved speaker style modeling. These studies demonstrate LoRA's effectiveness in capturing expressive speech features. However, its compositional potential remains underexplored, especially in combining LoRA modules to capture diverse, fine-grained speech attributes. To address this, we introduce TTS-Hub, a modular framework that trains LoRA modules for individual speech attributes and combines them at inference time, enabling flexible, interpretable user-controlled speech generation.

3 Proposed Method

In this section we present TTS-Hub, a novel framework for speech synthesis that enables fine-grained and flexible control through the composition of modular LoRA adapters. Our framework consists of three stages: (A) **Hub Construction** builds a data hub with labeled attributes and trains attribute-specific LoRA modules; (B) **LoRA Matching** selects and composes modules based on customized user intent; (C) **Inference with LoRA Adapters** injects the composite adapter into a frozen backbone to synthesize speech that aligns with the target features.

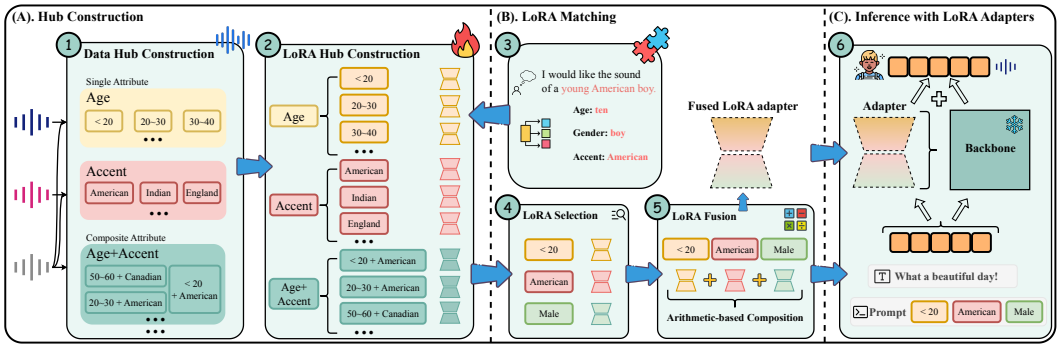


Figure 2: Overview of Our TTS-Hub.

3.1 Hub Construction

We first build the Data Hub by collecting and reorganizing samples from multiple public TTS datasets to ensure diverse speech attribute coverage. Then, we use attribute-specific subsets to train modular LoRA adapters, forming the LoRA Hub.

Data Hub. We collect multiple publicly available TTS datasets and organize a unified attribute taxonomy $A = A_a \cup A_c$, where:

- Atomic Attributes A_a. Each atomic attribute $a_j \in A_a, j = 1, \ldots, J$ represents an indivisible speaker characteristic, such as *"British accent"* or *"over 60 years old"*. J denotes the total number of atomic attributes. These attributes are grouped into attribute categories like *Accent*, *Age*, etc., enabling control over specific aspects of speech.
- Composite Attributes A_c. Each composite attribute (denoted as $\hat{a}_k \in A_c, k = 1, \ldots, K$) is a combination of two or more atomic attributes drawn from different categories. Formally, we write $\hat{a}_k = a_{k_1} + a_{k_2} \cdots + a_{k_n}$, where $a_{k_i} \in A_a$, $n \geq 2$ is the number of atomic attributes involved in $\hat{a}_k$, and K is the number of composite attributes. For example, the

composite attribute *"age: 20+accent: American"* merges the atomic attributes *"20 years old"* and *"American accent"*, forming a richer and more specific speaker profile.

For each atomic and composite attribute $a \in A$, we collect an attribute-specific subset $\mathcal{D}_a = \{x_i, y_i, a\}_{i=1}^{n_a}$ from the overall corpus, where x_i is the input text, y_i is the corresponding speech waveform, and a is the associated attribute annotation. The number of samples in $\mathcal{D}_a$ is denoted by n_a. The full Data Hub is therefore represented as:

$$\mathcal{D} = \{\mathcal{D}_a \mid a \in A\} \tag{1}$$

LoRA Hub. Given the data hub $\mathcal{D}$, we train a set of attribute-specialized LoRA adapters, collectively forming the LoRA Hub $\mathcal{H}$. Each adapter in $\mathcal{H}$ is trained to capture the effect of a specific attribute a, while keeping the backbone model $\mathcal{M}$ frozen. Let $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ denote a trainable weight matrix in $\mathcal{M}$ (e.g., a linear projection in the self-attention block), LoRA introduces a learnable low-rank update ΔW_a, applied additively to the frozen weight:

$$W_{\text{adapted}} = W + \Delta W_a, \quad \Delta W_a = \mathbf{B}_a \mathbf{A}_a \tag{2}$$

where $\mathbf{A}_a \in \mathbb{R}^{r \times d_{\text{in}}}$, $\mathbf{B}_a \in \mathbb{R}^{d_{\text{out}} \times r}$, and $r \ll \min(d_{\text{in}}, d_{\text{out}})$ is the bottleneck rank that controls the parameter efficiency. Only $\mathbf{A}$ and $\mathbf{B}$ are trainable, while the original backbone parameters W remain frozen. Since LoRA may be applied to multiple layers within the backbone model, we represent the complete adapter for attribute a as $\mathbf{\Delta}_a = \left\{\Delta W_a^{(l)} = \mathbf{B}_a^{(l)} \mathbf{A}_a^{(l)}\right\}_{l \in \mathcal{M}}$. The optimization objective is defined as:

$$\min_{\mathbf{\Delta}_a} \sum_{(x_i, y_i) \in \mathcal{D}_a} \mathcal{L}\left(\mathcal{M}_{\mathbf{\Delta}_a}(x_i),\ y_i\right) \tag{3}$$

where $\mathcal{L}$ is the training loss adopted by the backbone model $\mathcal{M}$. Each attribute $a \in A$ is associated with a subset $\mathcal{D}_a \subset \mathcal{D}$, which is used to train its corresponding adapter independently. Finally, we obtain the LoRA hub:

$$\mathcal{H} = \{\mathbf{\Delta}_a \mid a \in A\} \tag{4}$$

3.2 LoRA Matching

Once the LoRA Hub $\mathcal{H}$ is constructed, it enables flexible control over speech synthesis through modular selection and composition.

1) Single-attribute control. A specific attribute $a \in A$ can be controlled by directly injecting its corresponding LoRA adapter $\mathbf{\Delta}_a$ into the backbone model.

2) Multi-attribute control. Beyond individual attributes, more complex attribute behaviors can be expressed by composing multiple LoRA adapters. For instance, additive composition $\mathbf{\Delta}_{a_1} + \mathbf{\Delta}_{a_2} + \cdots + \mathbf{\Delta}_{a_n}$ allows simultaneous control over multiple attributes. Alternatively, subtractive operations can be used to remove a specific attribute from a composite adapter: $\mathbf{\Delta}_{a_{k_1} + a_{k_2} \cdots + a_{k_n}} - \mathbf{\Delta}_{a_{k_j}}$, where $\mathbf{\Delta}_{a_{k_1} + a_{k_2} \cdots + a_{k_n}}$ is a LoRA adapter for composite attribute $a_{k_1} + a_{k_2} \cdots + a_{k_n}$, and a_{k_j} is the attribute we would like to remove from the speech.

3) Attribute scaling. Furthermore, the strength of a single attribute can be adjusted by scaling the adapter with a weight w, as in $w \cdot \mathbf{\Delta}_a$, allowing gradual amplification or attenuation of the target attribute.

3.3 Inference with LoRA Adapters

The selected or composite LoRA adapter $\mathbf{\Delta}$ is integrated into the frozen backbone model $\mathcal{M}$. Given an input text x, the final output is synthesized as:

$$y' = \mathcal{M}_{\mathbf{\Delta}}(x) \tag{5}$$

In this way, our method eliminates the need for retraining the model for each attribute combination and enables flexible and interpretable control over speech attributes via arithmetic composition.

3.4 Framework Analysis

***Take-away*1. The framework's modular design provides unparalleled flexibility and scalability.** The use of modular LoRA adapters enables flexible composition of various attributes, allowing users to create novel voice styles without retraining. Additionally, new attributes can be easily integrated into the system by fine-tuning and adding new LoRA modules, ensuring seamless expansion without altering the backbone model.

***Take-away*2. The framework combines transparency with user control for more informed customization.** Unlike conventional black-box approaches, our framework offers interpretable control by explicitly composing LoRA modules, each dedicated to a specific attribute. This modular design enhances transparency, enabling users to easily trace and influence each attribute's contribution to the synthesized speech, while providing fine-grained customization options.

4 Experimental Settings

4.1 TTS-Hub Construction

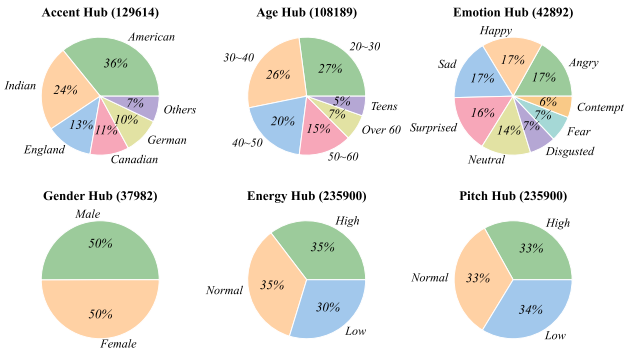


Figure 3: Attribute distributions in the Data Hub.

Data Hub Assembly. To support fine-grained controllable text-to-speech (TTS) using attribute-specific LoRA module composition, we reconstruct a large-scale speech dataset through a label-sharing strategy, integrating data from four publicly available corpora: Common Voice (Ardila et al., 2020), L2-ARCTIC (Zhao et al., 2018), Indic TTS (Baby et al., 2016), and TextrolSpeech (Ji et al., 2024b). Each utterance in the dataset is annotated with one or more fine-grained attribute labels. Specifically, the dataset covers six high-level attribute categories: Accent, Age, Emotion, Gender, Energy, and Pitch, comprising a total of 32 fine-grained subclasses. These include 10 accent types (e.g., American, Indian, British, Scottish), 6 age groups (ranging from under 20 years old to over 60 years old), 8 emotional states (e.g., Angry, Happy, Neutral), 2 gender classes (Male and Female), and 3 levels each for energy and pitch (High, Normal, and Low). In addition, based on label co-occurrence, we construct 9 sets of composite attribute data (e.g., American + over 60 years old) to support the subtractive validation experiments. We split the entire dataset into training, validation, and test sets using an 8:1:1 ratio. This carefully curated and diverse dataset provides a robust foundation for evaluating the fine-grained attribute controllability of TTS systems. The detailed distribution of the dataset is shown in Figure 3.

LoRA Hub Training. We adopt two backbone models: Parler-TTS (Lacombe et al., 2024) and VoiceLDM (Lee et al., 2024). We report results on Parler-TTS in the main paper, and leave the results on VoiceLDM to the supplementary materials due to space constraints. The former is based on a Transformer architecture, while the latter employs a diffusion-based architecture. Both frameworks support controllable speech synthesis purely through textual prompts. All experiments are conducted on a Linux system equipped with eight NVIDIA A40 GPUs, each featuring 48 GB memory. During training, we set the batch size to 8 and train for 10 epochs to ensure stable model convergence. Details of additional hyperparameter configurations are provided in the supplementary materials. We perform LoRA fine-tuning using all 32 single-attribute datasets. For the two key LoRA hyperparameters, namely LoRA Rank (r) and LoRA Alpha (α), we follow the configuration described in (Hu et al., 2022), and further validate the effectiveness of our chosen hyperparameters through additional experiments presented in the supplementary materials. Therefore, in all subsequent experiments, we consistently set r to 16 and α to 32 throughout the entire model training process. The process leading to this choice is elaborated in Appendix 6.

4.2 Evaluation Metrics

In order to comprehensively evaluate the performance of the synthesized speech, we first employ a human evaluation metric the Attribute Mean Opinion Score (**AMOS**), which assesses the perceptual expressiveness of LoRA modules on specific attributes from a human auditory perception perspective. The AMOS evaluation is conducted by a panel of 30 expert raters, all with research backgrounds in speech synthesis. Each participant is instructed to rate the quality of the synthesized audio on a scale from 1 to 5. Raters are compensated based on the number of evaluations they complete.

Moreover, we introduce two large language model-based metrics: **Win-Attr**, which quantifies the win rate in terms of attribute expressiveness, and **Win-Qual**, which assesses the win rate for overall speech quality. These metrics are derived through pairwise comparisons conducted with GPT-4o-Audio-Preview, which accepts audio inputs and returns final judgments in textual form. We employ two carefully designed prompts tailored to guide the model in comparing LoRA-based synthesis with natural language prompting, determining which method produces speech with superior quality or more salient attribute expression. In cases where the model finds the two audio samples comparable and unable to decisively differentiate them, it is allowed to declare a tie. The prompt details used in these evaluations are provided in Appendix 6.

In addition, three automatic objective metrics are adopted: Mel-Cepstral Distortion (**MCD**, (Kubichek, 1993)), F0 Root Mean Square Error (**F0-RMSE**, (Toda et al., 2007)), and Structural Similarity Index (**SSIM**, (Wang et al., 2004)), which measure the similarity between synthesized speech and ground truth.

4.3 Evaluation Rationale

Data Selection and Evaluation Based on Attribute Co-occurrence Given the scale of the training data, we operate under the hypothesis that data samples sharing a specific attribute will cluster together in the feature space, thereby sufficiently capturing the characteristic patterns of that attribute. To validate the model's capability in modeling individual attributes or their combinations, our experimental methodology is based on the principle of attribute label co-occurrence for both training and evaluation. Specifically, when conducting experiments targeting a particular attribute (e.g., "age=30"), we construct the dataset by selecting all samples annotated with the target attribute label. While these samples naturally exhibit variation in other attributes (such as emotion or accent), their shared target attribute is presumed to reflect the common feature characteristics relevant to it. This approach is necessitated by the practical constraint that real-world data rarely, if ever, contains samples that are identical across all attributes.

For the quantitative evaluation, we assess the performance of the synthesized speech by measuring its similarity to genuine samples possessing the target attribute. From the test set containing the attribute of interest, 100 samples are randomly selected to serve as the Ground Truth (GT). The similarity between each synthesized speech sample and this set of 100 GT samples is computed using relevant acoustic similarity metrics. The average of these scores is reported as the final measure of how well the synthesized speech captures the target attribute relative to the natural data. This evaluation protocol ensures a robust assessment focused on the attribute-specific (common characteristics), while accommodating the inherent diversity of other attributes in real-world data.

5 Results and Discussion

To evaluate the effectiveness of Modular LoRAs and their arithmetic composition for controlling speech synthesis, we first adopt individual LoRA adapters to examine whether single adapter can successfully model attribute-specific speech characteristics. We then explore three composition strategies: *addition*, *subtraction*, and *scaling*, which allow us to flexibly combine or modify attribute effects without retraining. These composition mechanisms facilitate modular and interpretable control over synthesized speech. An overview of the evaluation settings is presented in Table 1.

Table 1: Overview of experimental designs.

Control Type	Composition Strategy	Results	Control Type	Composition Strategy	Results
Single Attribute	$\mathbf{\Delta}_a$	Section 5.1	Addition	$w_1 \cdot \mathbf{\Delta}_{a_1} + w_2 \cdot \mathbf{\Delta}_{a_2}$	Section 5.2
Subtraction	$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	Section 5.3	Scaling	$w \cdot \mathbf{\Delta}_a$	Section 5.4

Table 2: Comparison of our TTS-Hub with ground truth (GT) and a natural-language prompting baseline (Prompt) under the single-attribute control setting. ↑ indicates metrics where higher values are better, and ↓ indicates metrics where lower values are preferred. Values in parentheses denote variances. *Note:* All results are reported based on the Parler-TTS model. Results on the alternative backbone VoiceLDM are included in the appendix and show consistent conclusions.

Attribute			AMOS↑	Win-Attr↑	Win-Qual↑	MCD↓	F0-RMSE↓	SSIM↑
Accent	Indian	GT	3.890(±0.0230)	-	-	-	-	-
		Prompt	2.920(±0.0433)	0.230(±0.0010)	0.368(±0.0030)	7.060	66.798	0.277
		Ours	3.750(±0.0050)	0.638(±0.0028)	0.550(±0.0020)	6.086	57.935	0.374
	US	GT	3.776(±0.0041)	-	-	-	-	-
		Prompt	3.732(±0.0029)	0.282(±0.0026)	0.452(±0.0024)	7.522	55.722	0.334
		Ours	3.736(±0.0077)	0.664(±0.0014)	0.446(±0.0005)	7.282	51.677	0.362
	England	GT	3.720(±0.0022)	-	-	-	-	-
		Prompt	3.510(±0.0182)	0.282(±0.0029)	0.354(±0.0049)	7.643	65.963	0.329
		Ours	3.770(±0.0108)	0.704(±0.0020)	0.456(±0.0056)	7.252	53.823	0.377
	Canadian	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.090(±0.0380)	0.262(±0.0049)	0.338(±0.0017)	7.590	66.140	0.341
		Ours	3.614(±0.0080)	0.620(±0.0037)	0.458(±0.0045)	7.138	51.251	0.407
Age	<20	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	2.974(±0.0109)	0.148(±0.0016)	0.378(±0.0052)	8.328	87.401	0.311
		Ours	3.754(±0.0098)	0.666(±0.0032)	0.378(±0.0032)	7.484	77.262	0.348
	20-29	GT	3.580(±0.0133)	-	-	-	-	-
		Prompt	3.120(±0.0158)	0.270(±0.0072)	0.488(±0.0020)	8.408	80.402	0.310
		Ours	3.390(±0.0080)	0.658(±0.0051)	0.438(±0.0023)	7.389	73.073	0.338
	30-39	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.400(±0.0213)	0.260(±0.0017)	0.596(±0.0039)	7.714	62.773	0.328
		Ours	3.430(±0.0095)	0.626(±0.0060)	0.386(±0.0019)	7.412	50.999	0.374
	>60	GT	3.906(±0.0107)	-	-	-	-	-
		Prompt	2.990(±0.0180)	0.240(±0.0034)	0.340(±0.0015)	7.250	66.951	0.325
		Ours	3.744(±0.0049)	0.698(±0.0021)	0.536(±0.0029)	7.585	58.022	0.373
Overall		GT	3.790±(0.0102)	-	-	-	-	-
		Prompt	3.344±(0.0196)	0.254(±0.0038)	0.376(±0.0061)	7.330	66.429	0.314
		Ours	3.661±(0.0102)	0.600(±0.0051)	0.409(±0.0079)	6.953	57.800	0.363

5.1 Evaluation under Single-Attribute Control

LoRA-based control significantly outperforms Prompt-only methods. We implement single-attribute controllable synthesis by prompting the backbone model equipped with attribute-specific LoRA modules. Due to the large number of attributes, we sample 16 audio clips per attribute for human evaluation. All other metrics are computed over the entire test set. To ensure the stability of the Large Language Model-based metrics, each pairwise comparison is evaluated five times. As shown in Table 2, the LoRA-based approach consistently outperforms prompt-based control across multiple evaluation metrics for the vast majority of attributes. This advantage is particularly evident in two metrics specifically designed to assess attribute expressiveness. For instance, for the highly distinguishable attribute "age over 60", our method significantly surpasses the prompt-based approach, improving AMOS from 2.990 to 3.744 and Win-Attr from 0.240 to 0.698. Although in some cases, such as the "US accent" attribute, our approach shows a slight drop in Win-Qual compared to the prompt-based method, it still demonstrates a clear advantage in attribute expressiveness, improving Win-Attr from 0.282 to 0.664. These results underscore the effectiveness of LoRA modules in capturing fine-grained acoustic characteristics. Furthermore, to mitigate potential bias stemming from the distinctiveness of individual attributes, we construct an attribute-invariant test set by randomly sampling 128 utterances from all single-attribute generations. As shown in the last three rows

Table 3: AMOS scores with varying weights between two LoRA modules. AMOS-a_1 and AMOS-a_2 reflect the perceived strength of attributes a_1 and a_2, with higher scores indicating stronger expression of the target attribute.

Composition	AMOS-a_1 ↑	AMOS-a_2 ↑	Composition	AMOS-a_1 ↑	AMOS-a_2 ↑
$0.0 \cdot \mathbf{\Delta}_{a_1} + 1.0 \cdot \mathbf{\Delta}_{a_2}$	1.803(±0.0577)	3.717(±0.0084)	$0.1 \cdot \mathbf{\Delta}_{a_1} + 0.9 \cdot \mathbf{\Delta}_{a_2}$	1.761(±0.0124)	3.644(±0.0163)
$0.2 \cdot \mathbf{\Delta}_{a_1} + 0.8 \cdot \mathbf{\Delta}_{a_2}$	1.834(±0.0263)	3.549(±0.0312)	$0.3 \cdot \mathbf{\Delta}_{a_1} + 0.7 \cdot \mathbf{\Delta}_{a_2}$	2.409(±0.0875)	3.456(±0.0177)
$0.4 \cdot \mathbf{\Delta}_{a_1} + 0.6 \cdot \mathbf{\Delta}_{a_2}$	2.419(±0.0484)	3.126(±0.0434)	$0.6 \cdot \mathbf{\Delta}_{a_1} + 0.4 \cdot \mathbf{\Delta}_{a_2}$	3.185(±0.0291)	2.761(±0.0462)
$0.7 \cdot \mathbf{\Delta}_{a_1} + 0.3 \cdot \mathbf{\Delta}_{a_2}$	3.387(±0.0182)	2.337(±0.0507)	$0.8 \cdot \mathbf{\Delta}_{a_1} + 0.2 \cdot \mathbf{\Delta}_{a_2}$	3.565(±0.0292)	2.323(±0.0374)
$0.9 \cdot \mathbf{\Delta}_{a_1} + 0.1 \cdot \mathbf{\Delta}_{a_2}$	3.619(±0.0292)	1.874(±0.0646)	$1.0 \cdot \mathbf{\Delta}_{a_1} + 0.0 \cdot \mathbf{\Delta}_{a_2}$	3.832(±0.0186)	1.383(±0.0607)

Table 4: Performance of multi-attribute control. #Attr denotes the number of composite attributes.

Method	#Attr	AMOS↑	Win-Attr↑	Win-Qual↑	#Attr	AMOS↑	Win-Attr↑	Win-Qual↑
Prompt	1	3.344 (± 0.0196)	0.254 (± 0.0038)	0.376 (± 0.0061)	2	2.937 (± 0.0452)	0.214 (± 0.0013)	0.367 (± 0.0080)
Ours		3.661 (± 0.0102)	0.600 (± 0.0051)	0.409 (± 0.0079)		3.625 (± 0.0077)	0.594 (± 0.0118)	0.408 (± 0.0074)
Prompt	3	2.656 (± 0.0529)	0.217 (± 0.0018)	0.333 (± 0.0051)	4	2.196 (± 0.0425)	0.296 (± 0.0030)	0.337 (± 0.0020)
Ours		3.281 (± 0.0322)	0.604 (± 0.0152)	0.379 (± 0.0048)		2.525 (± 0.0625)	0.397 (± 0.0039)	0.362 (± 0.0016)

of Table 2, the overall performance demonstrates that our approach consistently achieves superior attribute expressiveness.

5.2 Evaluation under Multiple-Attribute Control

Two-Attribute Control. We investigate the compositional behavior of fine-grained attributes under LoRA-based control by combining two specific attributes: accent (a_1) and (a_2). In total, we conduct four groups of fine-grained attribute combination experiments. The corresponding LoRA modules are fused through weighted summation, with weights ranging from 0.1 to 0.9 in increments of 0.2, to examine the impact of fine-grained weighting on synthesis quality. We randomly sample 128 audio clips from the combined set of all four groups for evaluation. As shown in Table 3, the results reveal a clear trade-off between the two attributes: increasing the weight of a_1 consistently improves the AMOS-a_1 score, while AMOS-a_2 gradually declines. This trend highlights the controllability and flexibility of our approach, enabling smooth and continuous transitions between multiple attribute representations. Moreover, the disentangled behavior suggests that LoRA modules effectively preserve attribute-specific information, thereby better supporting compositional control across features.

Multi-Attribute Control. In order to explore the upper limit of attribute composition using LoRA modules, we conducted experiments with an increased number of attributes, focusing specifically on combinations involving three and four features. Similarly, we mixed all multi-attribute combinations and randomly sampled 128 clips for evaluation. The same test set sampling strategy is applied for the subsequent subtractive and scaling experiments. The overall performance across different configurations is summarized in Table 4. The results indicate that composing three attributes with LoRA modules still yields strong synthesis performance, particularly in metrics closely related to attribute expressiveness such as AMOS and Win-Attr. However, when the number of attributes increases to four, we observe a notable degradation in synthesis quality. This finding suggests that introducing too many fused LoRA modules may impair the performance of the backbone model, which is consistent with the conclusions drawn in (Hu et al., 2022). Despite this limitation, the three-feature combination already covers a broad range of personalized synthesis scenarios.

5.3 Exploring Attribute Subtraction

To investigate whether specific attributes can be selectively attenuated from a composite LoRA representation, we conducted arithmetic subtraction experiments. Specifically, we subtracted an attribute-specific LoRA module (e.g., $\mathbf{\Delta}a_2$) from a composite module trained on a combined attribute set (e.g., $\mathbf{\Delta}a_1 + a_2$). This setup aims to assess whether subtraction can suppress the influence of the removed attribute while retaining the representation of the remaining one.

As shown in Table 5, subtracting $\mathbf{\Delta}_{a_2}$ from $\mathbf{\Delta}_{a_1+a_2}$ results in a noticeable decrease in AMOS for attribute a_2, dropping from 3.548 to 2.889, which indicates successful attenuation of the a_2-related representation. Meanwhile, we observe a degradation in AMOS for attribute a_1, decreasing from

3.615 to 3.317. This suggests that subtractive composition between LoRA modules may partially impair the original attribute expressiveness within the composite embedding. Additionally, a decline in overall Win-Qual scores is observed compared to the unmodified composite $\mathbf{\Delta}_{a_1+a_2}$, implying that such arithmetic operations may compromise the general synthesis quality. This trade-off highlights the inherent limitations of arithmetic operations within the LoRA parameter space to some extent.

Table 5: Evaluation of composition between two LoRA modules. AMOS-a_1 and AMOS-a_2 assess the presence of attributes a_1 and a_2, respectively. Win-Attr-a_1 and Win-Attr-a_2 indicate the pairwise win rates for correctly expressing a_1 and a_2. Win-Qual measures overall audio quality preference.

Subtraction	**AMOS-a_1 ↑**	**AMOS-a_2 ↑**	**Win-Attr-a_1 ↑**	**Win-Attr-a_2 ↑**	**Win-Qual↑**
GT	3.891(±0.0180)	3.813(±0.0206)	-	-	-
Prompt	3.270(±0.0114)	3.066(±0.0280)	0.276(±0.0036)	0.213(±0.0026)	0.390(±0.0052)
$\mathbf{\Delta}_{a_1+a_2}$	3.615(±0.0137)	3.548(±0.0249)	0.341(±0.0031)	0.342(±0.0050)	0.382(±0.0019)
$\mathbf{\Delta}_{a_1}$	3.450(±0.0220)	2.989(±0.1300)	0.338(±0.0030)	0.268(±0.0026)	0.364(±0.0032)
$\mathbf{\Delta}_{a_2}$	2.817(±0.0455)	3.478(±0.0352)	0.241(±0.0029)	0.375(±0.0025)	0.366(±0.0021)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	3.317(±0.0215)	2.889(±0.1549)	0.315(±0.0019)	0.278(±0.0022)	0.335(±0.0018)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	2.984(±0.0477)	3.494(±0.0417)	0.259(±0.0030)	0.353(±0.0030)	0.347(±0.0038)

5.4 Scaling Experiments on Individual Attributes

To further investigate whether LoRA modules exhibit scalable control over individual attributes, we conducted a series of scaling experiments using LoRA modules fine-tuned on single-attribute datasets. Module scaling involves applying a scalar coefficient to modulate the influence of an attribute-specific LoRA module on the backbone model.

Experimental results in Figure 4 reveal that when the scaling factor approaches zero, the target attribute's expressiveness is substantially attenuated. As the coefficient increases, the attribute becomes progressively more salient. However, beyond a certain threshold, further amplification yields diminishing perceptual gains, indicating a saturation effect in controllability. In additional, we examined the impact of negative scaling values. While negative coefficients reduced the prominence of the attribute to some extent, they did not induce an inverse or antagonistic acoustic effect. This suggests that the scaling mechanism based on LoRA modules indeed enables continuous and degree-controllable modulation of attribute strength, but lacks the ability to reverse the semantic polarity of attribute expression.

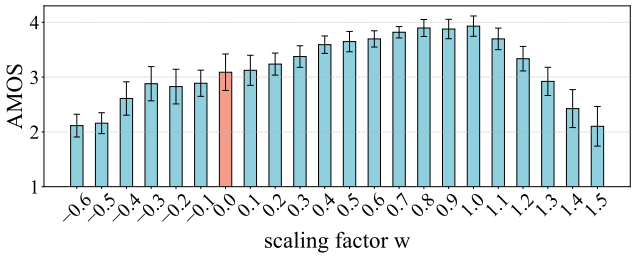


Figure 4: AMOS results under different scaling factors, where red denotes the absence of any attribute-specific LoRA modules.

6 Conclusion

This paper introduces TTS-Hub, a novel framework for controllable text-to-speech synthesis that leverages modular LoRA adapters and their arithmetic composition to enable fine-grained, flexible, and extensible control over speech attributes. By reconstructing an attribute-specific Data Hub and fine-tuning LoRA modules accordingly, we build a versatile LoRA Hub in which each adapter is specialized for a distinct speech attribute. During inference, TTS-Hub composes selected adapters through simple arithmetic operations, allowing intuitive multi-attribute control without retraining the backbone model. Experimental results show that individual adapters achieve precise control over single attributes, while their compositions support interpretable and robust multi-attribute synthesis. Furthermore, the modular architecture enables seamless integration of new attributes, highlighting the potential of LoRA-based modularity as a powerful and scalable paradigm for developing controllable TTS systems.

Ethics statement

Our research utilizes only open-source audio datasets for experimental purposes. All data have been thoroughly anonymized by their original providers, ensuring that no personal identifiers are present. As such, our use of these datasets does not involve any infringement of personal privacy.

Furthermore, the speech synthesis framework proposed in this study is intended solely for academic and scientific research. Any application of this technology must comply with legal and ethical standards. We explicitly prohibit its use for unlawful activities or any purpose that may cause harm to individuals or society.

Reproducibility statement

This study is committed to ensuring the reproducibility of its findings. To this end, we have ensured full transparency regarding data, methodology, and implementation. All experiments are based on publicly accessible datasets, which are detailed in Section 4.1. A complete description of the model architectures, evaluation metrics, and experimental procedures is provided in Section 4.1, 4.2 and 4.3. Furthermore, the complete source code, including configurations for training and inference, is available in the supplementary materials to facilitate the replication of our results.

References

Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber. Common voice: A massively-multilingual speech corpus. In *Proceedings of The 12th Language Resources and Evaluation Conference LREC*, pp. 4218–4222. European Language Resources Association, 2020.

Arun Baby, Anju Leela Thomas, NL Nishanthi, TTS Consortium, et al. Resources for indian languages. In *Proceedings of Text, Speech and Dialogue*, 2016.

Huda Barakat, Oytun Türk, and Cenk Demiroglu. Deep learning-based expressive speech synthesis: a systematic review of approaches, challenges, and resources. *EURASIP J. Audio Speech Music. Process.*, 2024(1):11, 2024. doi: 10.1186/S13636-024-00329-7. URL https://doi.org/10.1186/s13636-024-00329-7.

Lukasz Bondaruk and Jakub Kubiak. Lorp-tts: Low-rank personalized text-to-speech. *CoRR*, abs/2502.07562, 2025.

Thomas Bott, Florian Lux, and Ngoc Thang Vu. Controlling emotion in text-to-speech with natural language prompts. In *25th Annual Conference of the International Speech Communication Association, Interspeech ISCA*, 2024.

Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.

Zhihao Du, Qian Chen, Shiliang Zhang, Kai Hu, Heng Lu, Yexin Yang, Hangrui Hu, Siqi Zheng, Yue Gu, Ziyang Ma, Zhifu Gao, and Zhijie Yan. Cosyvoice: A scalable multilingual zero-shot text-to-speech synthesizer based on supervised semantic tokens. *CoRR*, abs/2407.05407, 2024a.

Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, Fan Yu, Huadai Liu, Zhengyan Sheng, Yue Gu, Chong Deng, Wen Wang, Shiliang Zhang, Zhijie Yan, and Jingren Zhou. Cosyvoice 2: Scalable streaming speech synthesis with large language models. *CoRR*, abs/2412.10117, 2024b. doi: 10.48550/ARXIV.2412.10117. URL https://doi.org/10.48550/arXiv.2412.10117.

Zhifang Guo, Yichong Leng, Yihan Wu, Sheng Zhao, and Xu Tan. Prompttts: Controllable text-to-speech with text descriptions. In *International Conference on Acoustics, Speech and Signal Processing ICASSP*, pp. 1–5. IEEE, 2023.

Bing Han, Long Zhou, Shujie Liu, Sanyuan Chen, Lingwei Meng, Yanming Qian, Yanqing Liu, Sheng Zhao, Jinyu Li, and Furu Wei. VALL-E R: robust and efficient zero-shot text-to-speech synthesis via monotonic alignment. *CoRR*, abs/2406.07855, 2024. doi: 10.48550/ARXIV.2406.07855. URL https://doi.org/10.48550/arXiv.2406.07855.

Hongkun Hao, Long Zhou, Shujie Liu, Jinyu Li, Shujie Hu, Rui Wang, and Furu Wei. Boosting large language model for speech synthesis: An empirical study. In *International Conference on Acoustics, Speech and Signal Processing ICASSP*, 2025.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *The Tenth International Conference on Learning Representations ICLR*. OpenReview.net, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Chengsong Huang, Qian Liu, Bill Yuchen Lin, Tianyu Pang, Chao Du, and Min Lin. Lorahub: Efficient cross-task generalization via dynamic lora composition, 2024a. URL https://arxiv.org/abs/2307.13269.

Rongjie Huang, Ruofan Hu, Yongqi Wang, Zehan Wang, Xize Cheng, Ziyue Jiang, Zhenhui Ye, Dongchao Yang, Luping Liu, Peng Gao, and Zhou Zhao. Instructspeech: Following speech editing instructions via large language models. In *Forty-first International Conference on Machine Learning ICML*. OpenReview.net, 2024b. URL https://openreview.net/forum?id=xlWcdtCyOC.

Rongjie Huang, Mingze Li, Dongchao Yang, Jiatong Shi, Xuankai Chang, Zhenhui Ye, Yuning Wu, Zhiqing Hong, Jiawei Huang, Jinglin Liu, Yi Ren, Yuexian Zou, Zhou Zhao, and Shinji Watanabe. Audiogpt: Understanding and generating speech, music, sound, and talking head. In *Thirty-Eighth AAAI Conference on Artificial Intelligence AAAI*, pp. 23802–23804. AAAI Press, 2024c.

Shengpeng Ji, Jialong Zuo, Minghui Fang, Ziyue Jiang, Feiyang Chen, Xinyu Duan, Baoxing Huai, and Zhou Zhao. Textrolspeech: A text style control speech corpus with codec language text-to-speech models. In *International Conference on Acoustics, Speech and Signal Processing ICASSP*, pp. 10301–10305. IEEE, 2024a. doi: 10.1109/ICASSP48485.2024.10445879.

Shengpeng Ji, Jialong Zuo, Minghui Fang, Ziyue Jiang, Feiyang Chen, Xinyu Duan, Baoxing Huai, and Zhou Zhao. Textrolspeech: A text style control speech corpus with codec language text-to-speech models. In *International Conference on Acoustics, Speech and Signal Processing, ICASSP*, pp. 10301–10305. IEEE, 2024b.

Shengpeng Ji, Qian Chen, Wen Wang, Jialong Zuo, Minghui Fang, Ziyue Jiang, Hai Huang, Zehan Wang, Xize Cheng, Siqi Zheng, and Zhou Zhao. ControlSpeech: Towards simultaneous and independent zero-shot speaker cloning and zero-shot language style control. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics ACL*, 2025.

Ziyue Jiang, Jinglin Liu, Yi Ren, Jinzheng He, Zhenhui Ye, Shengpeng Ji, Qian Yang, Chen Zhang, Pengfei Wei, Chunfeng Wang, Xiang Yin, Zejun Ma, and Zhou Zhao. Mega-tts 2: Boosting prompting mechanisms for zero-shot speech synthesis, 2024. URL https://arxiv.org/abs/2307.07218.

Zeqian Ju, Yuancheng Wang, Kai Shen, Xu Tan, Detai Xin, Dongchao Yang, Yanqing Liu, Yichong Leng, Kaitao Song, Siliang Tang, Zhizheng Wu, Tao Qin, Xiang-Yang Li, Wei Ye, Shikun Zhang, Jiang Bian, Lei He, Jinyu Li, and Sheng Zhao. Naturalspeech 3: Zero-shot speech synthesis with factorized codec and diffusion models, 2024. URL https://arxiv.org/abs/2403.03100.

Robert Kubichek. Mel-cepstral distance measure for objective speech quality assessment. In *Proceedings of IEEE pacific rim conference on communications computers and signal processing*, volume 1, pp. 125–128, 1993.

Yoach Lacombe, Vaibhav Srivastav, and Sanchit Gandhi. Parler-tts. https://github.com/huggingface/parler-tts, 2024.

Yeonghyeon Lee, Inmo Yeon, Juhan Nam, and Joon Son Chung. Voiceldm: Text-to-speech with environmental context. In *International Conference on Acoustics, Speech and Signal Processing, ICASSP*, pp. 12566–12571. IEEE, 2024.

Rui Liu, Berrak Sisman, Guanglai Gao, and Haizhou Li. Controllable accented text-to-speech synthesis with fine and coarse-grained intensity rendering. *IEEE ACM Trans. Audio Speech Lang. Process.*, 32:2188–2201, 2024.

Haowei Lou, Hye-Young Paik, Wen Hu, and Lina Yao. Stylespeech: Parameter-efficient fine tuning for pre-trained controllable text-to-speech. In *Proceedings of the 6th ACM International Conference on Multimedia in Asia MM*, pp. 3:1–3:7. ACM, 2024.

OpenAI. Gpt-4o mini tts. https://platform.openai.com/docs/models/gpt-4o-mini-tts, 2025. Accessed: 2025-08-02.

Xin Qi, Ruibo Fu, Zhengqi Wen, Jianhua Tao, Shuchen Shi, Yi Lu, Zhiyong Wang, Xiaopeng Wang, Yuankun Xie, Yukun Liu, Guanjun Li, Xuefei Liu, and Yongwei Li. EELE: exploring efficient and extensible lora integration in emotional text-to-speech. In *International Symposium on Chinese Spoken Language Processing ISCSLP*, pp. 294–298. IEEE, 2024.

Qwen. Qwen-tts: A versatile text-to-speech service. https://github.com/mco2004/qwen-tts, 2025. Accessed: 2025-08-02.

Yi Ren, Yangjun Ruan, Xu Tan, Tao Qin, Sheng Zhao, Zhou Zhao, and Tie-Yan Liu. Fastspeech: Fast, robust and controllable text to speech. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d'Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems NeurIPS*, pp. 3165–3174, 2019.

Giuseppe Ruggiero, Enrico Zovato, Luigi Di Caro, and Vincent Pollet. Voice cloning: a multi-speaker text-to-speech synthesis approach based on transfer learning. *CoRR*, abs/2102.05630, 2021.

Jonathan Shen, Ruoming Pang, Ron J. Weiss, Mike Schuster, Navdeep Jaitly, Zongheng Yang, Zhifeng Chen, Yu Zhang, Yuxuan Wang, R. J. Skerry-Ryan, Rif A. Saurous, Yannis Agiomyrgiannakis, and Yonghui Wu. Natural TTS synthesis by conditioning wavenet on MEL spectrogram predictions. In *International Conference on Acoustics, Speech and Signal Processing, ICASSP*, pp. 4779–4783. IEEE, 2018.

Xu Tan, Jiawei Chen, Haohe Liu, Jian Cong, Chen Zhang, Yanqing Liu, Xi Wang, Yichong Leng, Yuanhao Yi, Lei He, Sheng Zhao, Tao Qin, Frank K. Soong, and Tie-Yan Liu. Naturalspeech: End-to-end text-to-speech synthesis with human-level quality. *IEEE Trans. Pattern Anal. Mach. Intell.*, 46(6):4234–4245, 2024.

Tomoki Toda, Alan W Black, and Keiichi Tokuda. Voice conversion based on maximum-likelihood estimation of spectral parameter trajectory. *IEEE Transactions on Audio, Speech, and Language Processing*, 15(8):2222–2235, 2007.

Xinsheng Wang, Mingqi Jiang, Ziyang Ma, Ziyu Zhang, Songxiang Liu, Linqin Li, Zheng Liang, Qixi Zheng, Rui Wang, Xiaoqin Feng, Weizhen Bian, Zhen Ye, Sitong Cheng, Ruibin Yuan, Zhixian Zhao, Xinfa Zhu, Jiahao Pan, Liumeng Xue, Pengcheng Zhu, Yunlin Chen, Zhifei Li, Xie Chen, Lei Xie, Yike Guo, and Wei Xue. Spark-tts: An efficient llm-based text-to-speech model with single-stream decoupled speech tokens, 2025. URL https://arxiv.org/abs/2503.01710.

Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004.

Xun Wu, Shaohan Huang, and Furu Wei. Mixture of lora experts. In *The Twelfth International Conference on Learning Representations ICLR*. OpenReview.net, 2024.

Tianxin Xie, Yan Rong, Pengfei Zhang, and Li Liu. Towards controllable speech synthesis in the era of large language models: A survey. *CoRR*, abs/2412.06602, 2024. doi: 10.48550/ARXIV.2412.06602. URL https://doi.org/10.48550/arXiv.2412.06602.

Dongchao Yang, Songxiang Liu, Rongjie Huang, Chao Weng, and Helen Meng. Instructtts: Modelling expressive tts in discrete latent space with natural language style prompt. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32:2913–2925, 2024. doi: 10.1109/TASLP.2024.3402088.

Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and Jie Tang. Glm-4-voice: Towards intelligent and human-like end-to-end spoken chatbot. *CoRR*, abs/2412.02612, 2024.

Guanlong Zhao, Sinem Sonsaat, Alif Silpachai, Ivana Lucic, Evgeny Chukharev-Hudilainen, John Levis, and Ricardo Gutierrez-Osuna. L2-arctic: A non-native english speech corpus. In *Proc. Interspeech*, pp. 2783–2787, 2018. doi: 10.21437/Interspeech.2018-1110. URL http://dx.doi.org/10.21437/Interspeech.2018-1110.

THE USE OF LARGE LANGUAGE MODELS

Using an LLM to help with paper writing During the preparation of this work, the authors utilized Large Language Models for language polishing, improving the structural clarity of the manuscript, and refining the formal expression of individual sentences. The use of Large Language Models did not influence the substantive content of the study and served solely as a writing aid.

OVERALL OF THE APPENDIX

The supplementary material is organized into the following sections:

- Section A: Single-Attribute Control.
- Section B: Multiple-Attribute Control
- Section C: Fine-Grained Attributes Control
- Section D: Exploring Attribute Subtraction
- Section E: Exploring Attribute Scaling
- Section F: Prompts Utilized in the Evaluation Process and Hyperparameter Description
- Section J: Failure Modes Analysis
- Section H: Cosine Similarity Between all Adapter
- Section I: Visualization of AMOS Scores from Table 3, showing the impact of varying weights between two LoRA modules

A: SINGLE-ATTRIBUTE CONTROL

SINGLE-ATTRIBUTE CONTROL WITH PARLER-TTS

This section reports the complete experimental results for the single-attribute control setting, where we assess the effectiveness of LoRA-based control using the Parler-TTS model. We compare our proposed framework (TTS-Hub) against the ground truth (GT) and a natural language prompting baseline (Prompt). Metrics marked with $\uparrow$ indicate that higher values are better, while those marked with $\downarrow$ indicate that lower values are preferred. Values in parentheses denote the variance across samples. For detailed results, please refer to Table 7 and Table 8.

The experimental results demonstrate that our TTS-Hub exhibits significant advantages in fine-grained attribute control, providing strong evidence for the effectiveness of LoRA modules in learning and representing speech characteristics.

SINGLE-ATTRIBUTE CONTROL WITH VOICELDM

In this set of experiments, we adopt the diffusion-based VoiceLDM as the backbone model to evaluate single-attribute control. For detailed results, please refer to Table 9 and Table 10. The results demonstrate that our proposed TTS-Hub exhibits strong architectural robustness and can be effectively integrated with mainstream TTS frameworks.

B: MULTIPLE-ATTRIBUTE CONTROL

TWO-ATTRIBUTE CONTROL

To evaluate the arithmetic compositionality of LoRA modules, we conduct experiments by combining multiple pairs of LoRA modules, each corresponding to a distinct speech attribute. Similar to the single-attribute setting, we perform evaluations on both the Parler-TTS and VoiceLDM backbones. We report the performance of various attribute pair combinations. As shown in Table 11 and Table 12, combining two LoRA modules enables coherent and expressive speech synthesis in most cases, while maintaining effective control over each individual attribute. These results underscore the flexibility of modular attribute composition in LoRA-based TTS systems.

TRIPLE-ATTRIBUTE CONTROL

We investigate whether LoRA-based controllable generation can retain strong attribute expressiveness as the number of combined attributes increases by conducting triple-attribute control experiments on both backbone architectures. The results in Table 13 and Table 14 demonstrate that, while triple-attribute fusion generally preserves attribute expressiveness and intelligibility, a moderate degradation in overall synthesis quality is observed compared to the single- and two-attribute settings. This indicates that increasing the number of fused attributes may slightly compromise the synthesis quality of the backbone, though the performance remains within an acceptable range. These findings underscore the potential of LoRA modularity for controllable TTS synthesis across diverse and complex attribute configurations.

QUADRUPLE-ATTRIBUTE CONTROL

To further explore the upper limits of attribute composition under LoRA-based controllable synthesis, this set of experiments involves the fusion of four independently trained LoRA modules for simultaneous multi-attribute control in both Parler-TTS and VoiceLDM. Each configuration combines four distinct attributes to evaluate the scalability of modular LoRA-based generation.

As shown in Table 15 and Table 16, overall synthesis quality declines more noticeably compared to the fusion of one, two, or three attributes. While basic attribute expressiveness is preserved in some configurations, increased interference among modules often leads to reduced naturalness and stability. These findings reveal the limitations of naïve LoRA stacking at larger scales and underscore the need for more sophisticated fusion mechanisms or advanced composition strategies. Despite the observed degradation, this setting demonstrates the scalability potential of LoRA for fine-grained TTS control and motivates future research on controllable synthesis in more complex attribute spaces.

C: FINE-GRAINED ATTRIBUTES CONTROL

To investigate the impact of varying fusion weights assigned to different attributes during LoRA module composition, we conduct experiments using a finer-grained set of fusion weights in the two-attribute control setting. The results are presented in Tables 17a, 17b, 18a and 18b.

The findings show that increasing the weight of a specific attribute leads to a more pronounced manifestation of that attribute in the synthesized speech. Conversely, reducing the corresponding weight diminishes its expressiveness. These results confirm that LoRA-based fusion maintains strong controllability even at a fine-grained level, demonstrating its potential as a promising solution for precise and flexible voice editing tasks.

D: Exploring Attribute Subtraction

To explore whether specific speech attributes can be attenuated in the synthesized output through subtractive operations between LoRA modules, we conduct experiments on four attribute combinations. The goal is to assess whether such subtractive composition can support more flexible and targeted speech editing. Please refer to Tables 19, 20, 21, 22for detailed results.

The results indicate that subtractive operations are indeed effective in weakening the influence of targeted attributes in the output speech. However, this manipulation also introduces collateral effects, occasionally diminishing the expressiveness of remaining desired attributes or affecting overall speech naturalness. These findings suggest that while attribute subtraction via LoRA modules provides a viable pathway for fine-grained semantic control, it also reveals the inherent entanglement between attributes in the learned representation. Please see Table 23.

E: Exploring Attribute Scaling

In this set of experiments, we investigate whether scaling operations applied to individual LoRA modules can achieve smooth and continuous modulation of specific speech attributes. To this end, we select four prominent attributes and conduct controlled scaling experiments on their corresponding LoRA modules. The results are presented in Fig. 5a, 5b, 5c, and 5d.

The findings reveal three key observations. First, attribute expressiveness can be modulated in a relatively smooth and continuous manner by scaling the weight coefficients of the corresponding LoRA modules. Second, when the scaling factor exceeds a certain threshold, the overall speech quality deteriorates significantly. Third, applying negative scaling coefficients does not result in an inversion of the target attribute, contrary to initial expectations.

These results highlight the strong potential of LoRA-based control for fine-grained speech editing, while also emphasizing the critical importance of carefully selecting appropriate scaling factors to ensure stable and interpretable attribute manipulation.

F: Prompts Utilized in the Evaluation Process and Hyperparameter Description

When utilizing the GPT-4o-Audio API for speech evaluation, we design two distinct prompting strategies tailored to different evaluation dimensions.

For Win-Attr

LLM-Win-Qual Prompt:
You are a speech synthesis expert with extensive experience in evaluating audio quality from multiple perspectives.
We have developed two different text-to-speech (TTS) systems, and each system has generated one audio sample.
The order of the audio samples has been randomized (referred to as Audio Sample 1 and Audio Sample 2).
Please listen carefully to both samples and evaluate their overall audio quality.
Then, based on your analysis, indicate which sample (Audio Sample 1 or Audio Sample 2 or tie) you believe demonstrates higher overall quality.
If both samples are of comparable quality and you find it difficult to determine a clear winner, respond with“tie” instead of guessing.
Remember, only return me your final choice (Audio Sample 1 or Audio Sample 2 or tie)!

FOR WIN-QUAL

LLM-Win-Attr Prompt:
Assume you're now a speech expert with extensive experience in evaluating audio based on specific linguistic features, such as attribute.
You are presented with two audio clips, each generated by a different Text-to-Speech (TTS) system. The order of the clips has been randomized(referred to as Audio Sample 1 and Audio Sample 2), so you do not know which system produced which clip.
Your task is to carefully analyze both audio clips and assess how well each one conforms to the specified feature: attribute.
Focus on aspects such as pronunciation, intonation, and rhythm that are characteristic of the feature.
After your analysis, determine which audio clip (Audio Sample 1 or Audio Sample 2 or tie) better matches the specified feature and provide your conclusion.
If both samples are of comparable quality and you find it difficult to determine a clear winner, respond with"tie" instead of guessing.
Remember, only return me your final choice (Audio Sample 1 or Audio Sample 2 or tie)!

Parler-TTS Model We integrated LoRA modules into the decoder component of the Parler-TTS model. The ParlerTTSDecoder architecture comprises three main components: an `embed_tokens` layer, an `embed_positions` layer, and a 24-layer ParlerTTSDecoderLayer structure. Each ParlerTTSDecoderLayer contains self-attention (`self_attn`), encoder-attention (`encoder_attn`), activation function, and a two-layer feed-forward network (FC).

Specifically, LoRA was applied to the following components within each ParlerTTSDecoderLayer:

- Self-attention projections: `k_proj`, `q_proj`, `v_proj`, and `out_proj`
- Both fully connected layers in the feed-forward network

This integration enhances the trainable capacity of both self-attention mechanisms and feed-forward components.

VoiceLDM Model For the VoiceLDM model, LoRA was incorporated into the UNet module. The integration focuses on three critical components:

- Downsampling blocks: `CrossAttnDownBlock2D`
- Upsampling blocks: `CrossAttnUpBlock2D`
- Intermediate block: `UNetMidBlock2DCrossAttn`

Within these blocks, LoRA was applied to:

- Transformer query, key, and value projections
- Attention output projections
- Input and output convolutional layers
- Linear layers in FeedForward networks

Hyperparameter Configuration The key LoRA hyperparameters, rank (r) and scaling factor (α), were determined through systematic ablation studies as shown in Table 6. The optimal combination that yielded superior performance was selected for final model training.

LoRA Integration Specifications

Table 6: WER results of LoRA fine-tuned models with different (r, α) configurations compared to the base model.

Model / (r,α)	**(4,8)**	**(8,16)**	**(16,32)**	**(32,64)**	**(64,128)**	**(128,256)**
LoRA (Indian Accent) WER	17.55	25.02	13.50	20.27	13.13	13.06
LoRA (20 years old) WER	18.26	13.82	9.80	10.23	9.58	9.60
Base model WER	10.17	10.17	10.17	10.17	10.17	10.17

Table 7: Single LoRA Parler-tts table1

Metrics			**AMOS↑**	**Win-Attr↑**	**Win-Qual↑**	**MCD↓**	**F0-RMSE↓**	**SSIM↑**
Accent	India	GT	3.890(±0.0230)	-	-	-	-	-
		Prompt	2.920±(0.0433)	0.2 30(±0.0010)	0.368(±0.0030)	7.060	66.798	0.277
		Ours	3.750±(0.0050)	0.638(±0.0028)	0.550(±0.0020)	6.086	57.935	0.374
	US	GT	3.776(±0.0041)	-	-	-	-	-
		Prompt	3.732±(0.0029)	0.282(±0.0026)	0.452(±0.0024)	7.522	55.722	0.334
		Ours	3.736±(0.0077)	0.664(±0.0014)	0.446(±0.0005)	7.282	51.677	0.362
	England	GT	3.720(±0.0022)			-	-	-
		Prompt	3.510±(0.0182)	0.282(±0.0029)	0.354(±0.0049)	7.643	65.963	0.329
		Ours	3.770±(0.0108)	0.704(±0.0020)	0.456(±0.0056)	7.252	53.823	0.377
	Canadian	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.090±(0.0380)	0.262(±0.0049)	0.338(±0.0017)	7.590	66.140	0.341
		Ours	3.614±(0.0080)	0.620(±0.0037)	0.458(±0.0045)	7.138	51.251	0.407
	African	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.286±(0.0165)	0.250(±0.0010)	0.428(±0.0035)	8.537	74.751	0.301
		Ours	3.664±(0.0097)	0.660(±0.0004)	0.418(±0.0216)	7.850	65.504	0.328
	Australian	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.252±(0.0058)	0.222(±0.0020)	0.298(±0.0051)	8.008	74.914	0.335
		Ours	3.684±(0.0175)	0.640(±0.0015)	0.474(±0.0258)	7.454	69.470	0.376
	German	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.180±(0.0233)	0.250(±0.0067)	0.452(±0.0134)	7.931	76.689	0.284
		Ours	3.230±(0.0245)	0.570(±0.0050)	0.458(±0.0296)	7.303	71.360	0.454
	Irish	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.250±(0.0138)	0.308(±0.0031)	0.372(±0.0120)	7.002	59.462	0.360
		Ours	3.660±(0.0168)	0.610(±0.0067)	0.436(±0.0197)	5.954	23.537	0.519
	New Zealand	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.240±(0.0268)	0.182(±0.0012)	0.316(±0.0117)	7.814	75.104	0.321
		Ours	3.770±(0.0113)	0.650(±0.0017)	0.380(±0.0036)	7.107	70.368	0.381
	Scottish	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	2.990±(0.0305)	0.208(±0.0029)	0.464(±0.0043)	7.344	68.867	0.327
		Ours	3.632±(0.0083)	0.578(±0.0060)	0.414(±0.0081)	7.512	60.210	0.334
Age	<20	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	2.974±(0.0109)	0.148(±0.0016)	0.378(±0.0052)	8.328	87.401	0.311
		Ours	3.754±(0.0098)	0.666(±0.0032)	0.378(±0.0032)	7.484	77.262	0.348
	20-29	GT	3.580(±0.0133)	-	-	-	-	-
		Prompt	3.120±(0.0158)	0.270(±0.0072)	0.488(±0.0020)	8.408	80.402	0.310
		Ours	3.390±(0.0080)	0.658(±0.0051)	0.438(±0.0023)	7.389	73.073	0.338
	30-39	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.400±(0.0213)	0.260(±0.0017)	0.596(±0.0039)	7.714	62.773	0.328
		Ours	3.430±(0.0095)	0.626(±0.0060)	0.386(±0.0019)	7.412	50.999	0.374
	40-49	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.240±(0.0230)	0.228(±0.0028)	0.454(±0.0158)	7.792	77.185	0.306
		Ours	3.140±(0.0093)	0.552(±0.0071)	0.544(±0.0146)	7.533	59.221	0.378
	50-59	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.130±(0.0195)	0.244(±0.0040)	0.398(±0.0361)	7.330	70.847	0.313
		Ours	3.496±(0.0115)	0.618(±0.0091)	0.396(±0.0182)	7.368	63.911	0.379
	>60	GT	3.906(±0.0107)	-	-	-	-	-
		Prompt	2.990±(0.0180)	0.240(±0.0034)	0.340(±0.0015)	7.250	66.951	0.325
		Ours	3.744±(0.0049)	0.698(±0.0021)	0.536(±0.0029)	7.585	58.022	0.373

Table 8: Single LoRA Parler-tts table2

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑	MCD↓	F0-RMSE↓	SSIM↑
Emotion	Angry	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	2.880±(0.0570)	0.212(±0.0026)	0.314(±0.0017)	6.793	66.863	0.321
		Ours	3.306±(0.0057)	0.632(±0.0014)	0.362(±0.0051)	6.685	56.836	0.342
	Contempt	GT	3.580(±0.0133)	-	-	-	-	-
		Prompt	3.140±(0.0193)	0.236(±0.0051)	0.400(±0.0057)	7.055	55.141	0.276
		Ours	3.170±(0.0172)	0.586(±0.0054)	0.382(±0.0029)	6.399	47.880	0.308
	Disgusted	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.880±(0.0533)	0.222(±0.0028)	0.226(±0.0027)	7.547	73.899	0.209
		Ours	3.050±(0.0138)	0.566(±0.0097)	0.242(±0.0019)	7.027	51.770	0.275
	Fear	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.000±(0.0313)	0.430(±0.0016)	0.292(±0.0070)	7.301	73.298	0.209
		Ours	3.350±(0.0125)	0.400(±0.0062)	0.316(±0.0019)	7.041	72.044	0.261
	Happy	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.900±(0.0113)	0.234(±0.0023)	0.238(±0.0075)	6.976	79.335	0.293
		Ours	3.586±(0.0082)	0.648(±0.0018)	0.306(±0.0037)	6.564	71.381	0.384
	Neutral	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.700±(0.0015)	0.254(±0.0026)	0.306(±0.0074)	6.555	44.267	0.328
		Ours	3.698±(0.0060)	0.612(±0.0036)	0.312(±0.0025)	6.521	49.650	0.330
	Sad	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.670±(0.0145)	0.230(±0.0026)	0.316(±0.0056)	6.576	48.579	0.313
		Ours	3.728±(0.0035)	0.392(±0.0076)	0.386(±0.0065)	6.540	41.201	0.353
	Surprised	GT	3.906(±0.0107)	-	-	-	-	-
		Prompt	2.674±(0.0110)	0.200(±0.0007)	0.252(±0.0039)	6.843	96.548	0.292
		Ours	3.750±(0.0027)	0.444(±0.0041)	0.282(±0.0045)	6.709	89.640	0.346
Gender	Female	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	4.900±(0.0500)	0.340(±0.0118)	0.394(±0.0058)	7.992	58.487	0.330
		Ours	5.000±(0.0000)	0.590(±0.0134)	0.406(±0.0203)	7.288	53.530	0.387
	Male	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	5.000±(0.0000)	0.306(±0.0033)	0.440(±0.0025)	7.668	58.527	0.344
		Ours	4.900±(0.0500)	0.668(±0.0017)	0.464(±0.0024)	7.193	46.722	0.385
Energy	High	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.600±(0.0063)	0.264(±0.0068)	0.320(±0.0031)	6.930	67.648	0.341
		Ours	3.746±(0.0033)	0.570(±0.0097)	0.340(±0.0040)	6.871	50.762	0.356
	Normal	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.550±(0.0145)	0.284(±0.0062)	0.442(±0.0037)	6.577	57.822	0.349
		Ours	3.622±(0.0030)	0.574(±0.0022)	0.426(±0.0064)	6.512	54.277	0.354
	Low	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.702±(0.0019)	0.314(±0.0199)	0.394(±0.0057)	6.433	52.847	0.326
		Ours	3.676±(0.0031)	0.538(±0.0182)	0.418(±0.0006)	6.118	50.705	0.345
Pitch	High	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.806±(0.0172)	0.264(±0.0007)	0.486(±0.0013)	6.694	70.196	0.325
		Ours	3.786±(0.0017)	0.576(±0.0050)	0.448(±0.0047)	6.245	69.723	0.364
	Normal	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.530±(0.0057)	0.264(±0.0027)	0.424(±0.0020)	6.671	53.482	0.334
		Ours	3.580±(0.0180)	0.634(±0.0068)	0.412(±0.0068)	6.482	46.770	0.362
	Low	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.762±(0.0039)	0.202(±0.0012)	0.308(±0.0021)	6.666	38.810	0.366
		Ours	3.734±(0.0047)	0.618(±0.0018)	0.408(±0.0046)	6.604	39.073	0.375
Overall		GT	3.790±(0.0196)	-	-	-	-	-
		Prompt	3.344±(0.0196)	0.254(±0.0038)	0.377(±0.0061)	7.330	66.429	0.314
		Ours	3.661±(0.0102)	0.600(±0.0051)	0.409(±0.0076)	6.953	57.800	0.363

Table 9: Single LoRA VoiceLDM table1

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑	MCD↓	F0-RMSE↓	SSIM↑
Accent	India	GT	3.890(±0.0230)	-	-	-	-	-
		Prompt	2.979±(0.0525)	0.216±(0.0014)	0.350±(0.0061)	9.666	65.030	0.264
		Ours	3.391±(0.0102)	0.638±(0.0018)	0.566±(0.0020)	8.598	59.025	0.343
	US	GT	3.776(±0.0041)	-	-	-	-	-
		Prompt	3.625±(0.0119)	0.286±(0.0016)	0.424±(0.0041)	7.412	70.939	0.305
		Ours	3.663±(0.0035)	0.572±(0.0007)	0.440±(0.0006)	7.206	68.877	0.334
	England	GT	3.720(±0.0022)			-	-	-
		Prompt	3.192±(0.0327)	0.312±(0.0026)	0.336±(0.0073)	7.682	70.294	0.341
		Ours	3.734±(0.0297)	0.606±(0.0028)	0.422±(0.0036)	7.598	68.787	0.362
	Canadian	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.206±(0.0626)	0.268±(0.0013)	0.322±(0.0031)	7.608	101.237	0.349
		Ours	3.572±(0.0175)	0.500±(0.0084)	0.466±(0.0028)	7.250	80.957	0.391
	African	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.206±(0.0109)	0.302±(0.0033)	0.406±(0.0023)	8.039	81.730	0.251
		Ours	3.518±(0.0053)	0.556±(0.0019)	0.496±(0.0077)	7.335	74.216	0.309
	Australian	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.065±(0.0426)	0.270±(0.0068)	0.282±(0.0058)	7.406	56.671	0.358
		Ours	3.474±(0.0250)	0.556±(0.0019)	0.482±(0.0203)	7.035	50.224	0.393
	German	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	2.766±(0.0058)	0.272±(0.0063)	0.452±(0.0157)	7.828	61.770	0.303
		Ours	3.466±(0.0143)	0.502±(0.0054)	0.442±(0.0271)	7.412	56.454	0.366
	Irish	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.185±(0.0448)	0.330±(0.0049)	0.342±(0.0116)	7.872	112.735	0.320
		Ours	3.573±(0.0200)	0.550±(0.0053)	0.434±(0.0088)	7.103	87.870	0.465
	New Zealand	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.283±(0.0429)	0.234±(0.0073)	0.302±(0.0129)	7.111	84.131	0.261
		Ours	3.505±(0.0142)	0.610±(0.0055)	0.396±(0.0054)	6.729	75.663	0.311
	Scottish	GT	3.842(±0.0026)	-	-	-	-	-
		Prompt	3.248±(0.0556)	0.216±(0.0028)	0.444±(0.0091)	7.313	87.023	0.363
		Ours	3.286±(0.0226)	0.530±(0.0070)	0.432±(0.0119)	7.433	81.028	0.386
Age	<20	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	2.293±(0.0364)	0.148±(0.0016)	0.324±(0.0023)	8.526	91.606	0.210
		Ours	3.330±(0.0183)	0.604±(0.0013)	0.340±(0.0013)	7.812	82.244	0.285
	20-29	GT	3.580(±0.0133)	-	-	-	-	-
		Prompt	3.125±(0.0149)	0.270±(0.0055)	0.308±(0.0020)	7.773	80.275	0.250
		Ours	3.403±(0.0019)	0.580±(0.0024)	0.508±(0.0017)	7.393	65.687	0.264
	30-39	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.362±(0.0221)	0.304±(0.0028)	0.408±(0.0020)	7.959	55.910	0.269
		Ours	3.412±(0.0295)	0.510±(0.0051)	0.424±(0.0043)	7.659	50.485	0.313
	40-49	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.772±(0.0202)	0.258±(0.0029)	0.332±(0.0018)	7.778	62.638	0.260
		Ours	2.880±(0.0195)	0.494±(0.0041)	0.428±(0.0066)	7.635	59.207	0.283
	50-59	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.519±(0.0097)	0.274±(0.0036)	0.374±(0.0187)	7.780	58.911	0.274
		Ours	3.290±(0.0193)	0.558±(0.0061)	0.470±(0.0210)	7.601	53.792	0.297
	>60	GT	3.906(±0.0107)	-	-	-	-	-
		Prompt	2.338±(0.0268)	0.244±(0.0016)	0.280±(0.0005)	7.575	57.913	0.317
		Ours	3.273±(0.0201)	0.582±(0.0027)	0.504±(0.0082)	7.126	52.973	0.364

Table 10: Single LoRA VoiceLDM table2

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑	MCD↓	F0-RMSE↓	SSIM↑
Emotion	Angry	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	2.320±(0.0057)	0.216±(0.0023)	0.228±(0.0028)	8.744	80.619	0.294
		Ours	2.960±(0.0093)	0.542±(0.0011)	0.360±(0.0077)	8.041	74.315	0.339
	Contempt	GT	3.580(±0.0133)	-	-	-	-	-
		Prompt	3.110±(0.0055)	0.320±(0.0303)	0.380±(0.0337)	9.411	88.837	0.216
		Ours	3.090±(0.0068)	0.438±(0.0237)	0.398±(0.0174)	7.882	75.575	0.243
	Disgusted	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.460±(0.0730)	0.250±(0.0009)	0.378±(0.0095)	8.250	86.964	0.241
		Ours	2.780±(0.0220)	0.496±(0.0071)	0.388±(0.0193)	8.320	77.363	0.263
	Fear	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.860±(0.0168)	0.302±(0.0159)	0.416±(0.0297)	8.740	82.451	0.218
		Ours	3.410±(0.0155)	0.544±(0.0124)	0.440±(0.0105)	8.036	65.608	0.239
	Happy	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.370±(0.0395)	0.264±(0.0017)	0.360±(0.0122)	7.917	98.318	0.246
		Ours	3.170±(0.0545)	0.574±(0.0031)	0.464±(0.0102)	7.383	72.511	0.317
	Neutral	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	3.630±(0.0370)	0.298±(0.0029)	0.400±(0.0115)	6.690	59.095	0.286
		Ours	3.760±(0.0230)	0.486±(0.0072)	0.400±(0.0053)	6.770	65.419	0.282
	Sad	GT	3.500(±0.0063)	-	-	-	-	-
		Prompt	2.150±(0.0125)	0.312±(0.0277)	0.426±(0.0109)	7.772	74.204	0.224
		Ours	3.660±(0.0530)	0.506±(0.0152)	0.460±(0.0096)	7.562	75.800	0.253
	Surprised	GT	3.906(±0.0107)	-	-	-	-	-
		Prompt	2.180±(0.0470)	0.228±(0.0037)	0.284±(0.0016)	7.172	99.517	0.279
		Ours	3.480±(0.0220)	0.400±(0.0046)	0.346±(0.0013)	7.219	93.524	0.302
Gender	Female	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	5.000±(0.0000)	0.370±(0.0067)	0.400±(0.0055)	7.509	95.107	0.156
		Ours	5.000±(0.0000)	0.514±(0.0052)	0.408±(0.0042)	7.465	88.820	0.196
	Male	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	4.900±(0.0500)	0.316±(0.0027)	0.436±(0.0042)	7.984	66.466	0.183
		Ours	5.000±(0.0000)	0.570±(0.0009)	0.414±(0.0018)	6.708	50.520	0.287
Energy	High	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.530±(0.0070)	0.288±(0.0067)	0.364±(0.0046)	9.110	75.222	0.244
		Ours	3.760±(0.0092)	0.476±(0.0158)	0.426±(0.0211)	8.674	71.325	0.297
	Normal	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.620±(0.0058)	0.292±(0.0049)	0.352±(0.0025)	7.256	68.623	0.296
		Ours	3.620±(0.0033)	0.528±(0.0010)	0.402±(0.0041)	6.841	64.300	0.309
	Low	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.660±(0.0430)	0.324±(0.0183)	0.376±(0.0204)	9.483	86.203	0.265
		Ours	3.670±(0.0495)	0.460±(0.0088)	0.334±(0.0039)	9.127	69.790	0.286
Pitch	High	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.731±(0.0136)	0.270±(0.0015)	0.408±(0.0029)	7.392	80.564	0.140
		Ours	3.822±(0.0007)	0.520±(0.0008)	0.422±(0.0086)	6.657	73.766	0.195
	Normal	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.629±(0.0055)	0.280±(0.0037)	0.450±(0.0094)	5.954	54.105	0.205
		Ours	3.604±(0.0083)	0.558±(0.0029)	0.458±(0.0028)	5.492	52.942	0.239
	Low	GT	3.872(±0.0068)	-	-	-	-	-
		Prompt	3.611±(0.0016)	0.232±(0.0021)	0.292±(0.0017)	7.211	47.296	0.199
		Ours	3.657±(0.0068)	0.512±(0.0014)	0.338±(0.0047)	7.000	45.805	0.222
Overall		GT	3.795±(0.0145)	-	-	-	-	-
		Prompt	3.154±(0.0267)	0.274±(0.0059)	0.364±(0.0084)	7.873	76.325	0.262
		Ours	3.538±(0.0173)	0.534±(0.0054)	0.428±(0.0083)	7.441	68.277	0.304

Table 11: Dual LoRA Parler-tts table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
2 Attributes	UK+20	Prompt	2.900(±0.0613)	0.220(±0.0030)	0.296(±0.0048)
		Ours	3.730(±0.0108)	0.592(±0.0184)	0.322(±0.0030)
	UK+60	Prompt	2.880(±0.0133)	0.220(±0.0035)	0.382(±0.0033)
		Ours	3.668(±0.0124)	0.634(±0.0029)	0.430(±0.0245)
	UK+Happy	Prompt	2.848(±0.1334)	0.162(±0.0008)	0.322(±0.0071)
		Ours	3.620(±0.0066)	0.636(±0.0012)	0.378(±0.0028)
	UK+Sad	Prompt	2.742(±0.0391)	0.266(±0.0010)	0.398(±0.0036)
		Ours	3.720(±0.0060)	0.592(±0.0045)	0.370(±0.0017)
	UK+Surprised	Prompt	2.772(±0.0412)	0.194(±0.0002)	0.400(±0.0011)
		Ours	3.420(±0.0105)	0.638(±0.0010)	0.464(±0.0013)
	UK+teens	Prompt	2.792(±0.0196)	0.240(±0.0011)	0.414(±0.0088)
		Ours	3.550(±0.0062)	0.560(±0.0234)	0.462(±0.0027)
	US+20	Prompt	3.320(±0.0408)	0.212(±0.0008)	0.294(±0.0222)
		Ours	3.650(±0.0025)	0.622(±0.0016)	0.354(±0.0156)
	US+60	Prompt	3.030(±0.0208)	0.208(±0.0011)	0.412(±0.0053)
		Ours	3.620(±0.0110)	0.538(±0.0228)	0.448(±0.0020)
	US+teens	Prompt	3.150(±0.0375)	0.204(±0.0006)	0.382(±0.0158)
		Ours	3.648(±0.0037)	0.536(±0.0307)	0.448(±0.0133)
Overall		Prompt	2.937(±0.0452)	0.214(±0.0013)	0.367(±0.0080)
		Ours	3.625(±0.0077)	0.594(±0.0118)	0.408(±0.0074)

Table 12: Dual LoRA VoiceLDM table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
2 Attributes	UK+20	Prompt	2.780±(0.1145)	0.258±(0.0030)	0.258±(0.0053)
		Ours	3.570±(0.0208)	0.454±(0.0016)	0.358±(0.0025)
	UK+60	Prompt	2.660±(0.0168)	0.234±(0.0070)	0.368±(0.0091)
		Ours	3.740±(0.0668)	0.426±(0.0094)	0.416±(0.0055)
	UK+Happy	Prompt	2.918±(0.1915)	0.206±(0.0007)	0.308±(0.0103)
		Ours	3.350±(0.0195)	0.470±(0.0177)	0.368±(0.0015)
	UK+Sad	Prompt	2.820±(0.1407)	0.284±(0.0018)	0.380±(0.0049)
		Ours	3.608±(0.0494)	0.456±(0.0031)	0.376±(0.0010)
	UK+Surprised	Prompt	2.542±(0.0788)	0.204±(0.0010)	0.376±(0.0028)
		Ours	3.730±(0.1286)	0.430±(0.0183)	0.438±(0.0010)
	UK+teens	Prompt	3.010±(0.0330)	0.292±(0.0033)	0.394±(0.0088)
		Ours	3.624±(0.0261)	0.432±(0.0020)	0.472±(0.0067)
	US+20	Prompt	3.056±(0.0093)	0.224±(0.0011)	0.304±(0.0208)
		Ours	3.780±(0.0261)	0.388±(0.0056)	0.366±(0.0122)
	US+60	Prompt	2.760±(0.0318)	0.250±(0.0045)	0.410±(0.0022)
		Ours	3.255±(0.0146)	0.416±(0.0032)	0.428±(0.0032)
	US+teens	Prompt	2.895±(0.0240)	0.258±(0.0050)	0.366±(0.0220)
		Ours	3.554±(0.0161)	0.376±(0.0031)	0.412±(0.0024)
Overall		Prompt	2.827±(0.0711)	0.246±(0.0030)	0.352±(0.0096)
		Ours	3.579±(0.0409)	0.428±(0.0071)	0.404±(0.0040)

Table 13: Triple LoRA Parler-tts table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
3 Attributes	High Energy+Happy+US	Prompt	2.660(±0.0430)	0.224(±0.0024)	0.365(±0.0086)
		Ours	3.250(±0.0250)	0.594(±0.0128)	0.382(±0.0010)
	Low Pitch+Happy+UK	Prompt	2.520(±0.0970)	0.178(±0.0007)	0.376(±0.0099)
		Ours	3.220(±0.0333)	0.664(±0.0001)	0.434(±0.0102)
	60s+Happy+UK	Prompt	2.650(±0.0052)	0.236(±0.0007)	0.278(±0.0037)
		Ours	3.180(±0.0283)	0.652(±0.0081)	0.348(±0.0085)
	Teens+Happy+US	Prompt	2.870(±0.0108)	0.236(±0.0009)	0.302(±0.0016)
		Ours	3.354(±0.0081)	0.594(±0.0195)	0.404(±0.0011)
	Teens+Sad+UK	Prompt	2.618(±0.1514)	0.212(±0.0032)	0.310(±0.0047)
		Ours	3.352(±0.0490)	0.580(±0.0182)	0.322(±0.0054)
	Teens+Sad+US	Prompt	2.618(±0.0101)	0.214(±0.0028)	0.366(±0.0019)
		Ours	3.330(±0.0497)	0.538(±0.0325)	0.384(±0.0027)
Overall		Prompt	2.656(±0.0529)	0.217(±0.0018)	0.333(±0.0051)
		Ours	3.281(±0.0322)	0.604(±0.0152)	0.379(±0.0048)

Table 14: Triple LoRA VoiceLDM table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
3 Attributes	High Energy+Happy+US	Prompt	2.722±(0.0324)	0.294±(0.0301)	0.345±(0.0077)
		Ours	3.115±(0.0374)	0.474±(0.0199)	0.354±(0.0033)
	Low Pitch+Happy+UK	Prompt	2.350±(0.0413)	0.204±(0.0015)	0.366±(0.0055)
		Ours	2.881±(0.0117)	0.462±(0.0057)	0.422±(0.0010)
	60s+Happy+UK	Prompt	2.537±(0.0258)	0.258±(0.0006)	0.252±(0.0088)
		Ours	3.018±(0.1326)	0.516±(0.0007)	0.338±(0.0009)
	Teens+Happy+US	Prompt	3.052±(0.0074)	0.266±(0.0013)	0.310±(0.0008)
		Ours	3.201±(0.0048)	0.472±(0.0032)	0.376±(0.0004)
	Teens+Sad+UK	Prompt	2.653±(0.0755)	0.256±(0.0013)	0.304±(0.0074)
		Ours	3.008±(0.0089)	0.444±(0.0067)	0.384±(0.0030)
	Teens+Sad+US	Prompt	2.805±(0.0133)	0.238±(0.0013)	0.342±(0.0020)
		Ours	3.412±(0.0344)	0.396±(0.0025)	0.364±(0.0008)
Overall		Prompt	2.687±(0.0326)	0.253±(0.0060)	0.320±(0.0054)
		Ours	3.106±(0.0383)	0.461±(0.0065)	0.373±(0.0016)

Table 15: Quadra LoRA Parler-tts table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
4 Attributes	Teens+Happy+Indian+High Pitch	Prompt	2.200(±0.0288)	0.276(±0.0018)	0.374(±0.0012)
		Ours	2.424(±0.1037)	0.476(±0.0058)	0.356(±0.0016)
	Teens+Happy+UK+Low Pitch	Prompt	2.228(±0.0257)	0.322(±0.0058)	0.288(±0.0027)
		Ours	2.252(±0.0253)	0.414(±0.0047)	0.362(±0.0018)
	60s+Surprised+US+High Energy	Prompt	2.160(±0.0730)	0.290(±0.0016)	0.350(±0.0021)
		Ours	2.898(±0.0585)	0.300(±0.0013)	0.368(±0.0014)
Overall		Prompt	2.196(±0.0425)	0.296(±0.0030)	0.337(±0.0020)
		Ours	2.525(±0.0625)	0.397(±0.0039)	0.362(±0.0016)

Table 16: Quadra LoRA VoiceLDM table

Metrics			AMOS↑	Win-Attr↑	Win-Qual↑
4 Attributes	Teens+Happy+Indian+High Pitch	Prompt	2.070(±0.0145)	0.324(±0.0013)	0.316(±0.0023)
		Ours	2.190(±0.0753)	0.384(±0.0015)	0.410(±0.0116)
	Teens+Happy+UK+Low Pitch	Prompt	2.142(±0.0145)	0.332(±0.0088)	0.272(±0.0025)
		Ours	2.222(±0.0119)	0.360(±0.0018)	0.368(±0.0011)
	60s+Surprised+US+High Energy	Prompt	2.166(±0.0773)	0.282(±0.0032)	0.250(±0.0014)
		Ours	2.746(±0.0397)	0.330(±0.0017)	0.338(±0.0011)
Overall		Prompt	2.126(±0.0354)	0.313(±0.0044)	0.279(±0.0021)
		Ours	2.386(±0.0423)	0.358(±0.0017)	0.372(±0.0046)

Table 17: Weighted experiences across different attribute pairs.

Combination	AMOS-a_1 ↑	AMOS-a_2 ↑
$0.0 \cdot \mathbf{\Delta}_{a_1} + 1.0 \cdot \mathbf{\Delta}_{a_2}$	1.560(±0.1330)	3.740(±0.0090)
$0.1 \cdot \mathbf{\Delta}_{a_1} + 0.9 \cdot \mathbf{\Delta}_{a_2}$	1.226(±0.0234)	3.720(±0.0285)
$0.2 \cdot \mathbf{\Delta}_{a_1} + 0.8 \cdot \mathbf{\Delta}_{a_2}$	1.254(±0.0441)	3.684(±0.0117)
$0.3 \cdot \mathbf{\Delta}_{a_1} + 0.7 \cdot \mathbf{\Delta}_{a_2}$	2.120(±0.2570)	3.574(±0.0113)
$0.4 \cdot \mathbf{\Delta}_{a_1} + 0.6 \cdot \mathbf{\Delta}_{a_2}$	1.480(±0.0782)	3.292(±0.1106)
$0.5 \cdot \mathbf{\Delta}_{a_1} + 0.5 \cdot \mathbf{\Delta}_{a_2}$	3.028(±0.0465)	3.006(±0.0341)
$0.6 \cdot \mathbf{\Delta}_{a_1} + 0.4 \cdot \mathbf{\Delta}_{a_2}$	3.136(±0.0462)	2.988(±0.0342)
$0.7 \cdot \mathbf{\Delta}_{a_1} + 0.3 \cdot \mathbf{\Delta}_{a_2}$	3.290(±0.0092)	2.402(±0.0738)
$0.8 \cdot \mathbf{\Delta}_{a_1} + 0.2 \cdot \mathbf{\Delta}_{a_2}$	3.396(±0.0113)	2.276(±0.0513)
$0.9 \cdot \mathbf{\Delta}_{a_1} + 0.1 \cdot \mathbf{\Delta}_{a_2}$	3.636(±0.0212)	2.100(±0.0700)
$1.0 \cdot \mathbf{\Delta}_{a_1} + 0.0 \cdot \mathbf{\Delta}_{a_2}$	3.684(±0.0063)	1.310(±0.1330)

(a) Weighted experience: $\mathbf{\Delta}_{a_1}$: Indian, $\mathbf{\Delta}_{a_2}$: Happy

Combination	AMOS-a_1 ↑	AMOS-a_2 ↑
$0.0 \cdot \mathbf{\Delta}_{a_1} + 1.0 \cdot \mathbf{\Delta}_{a_2}$	1.190(±0.0430)	3.520(±0.0108)
$0.1 \cdot \mathbf{\Delta}_{a_1} + 0.9 \cdot \mathbf{\Delta}_{a_2}$	1.100(±0.0100)	3.402(±0.0099)
$0.2 \cdot \mathbf{\Delta}_{a_1} + 0.8 \cdot \mathbf{\Delta}_{a_2}$	1.250(±0.0325)	3.470(±0.0322)
$0.3 \cdot \mathbf{\Delta}_{a_1} + 0.7 \cdot \mathbf{\Delta}_{a_2}$	2.040(±0.0680)	3.344(±0.0409)
$0.4 \cdot \mathbf{\Delta}_{a_1} + 0.6 \cdot \mathbf{\Delta}_{a_2}$	2.350(±0.0550)	2.870(±0.0545)
$0.5 \cdot \mathbf{\Delta}_{a_1} + 0.5 \cdot \mathbf{\Delta}_{a_2}$	2.550(±0.1375)	2.550(±0.0213)
$0.6 \cdot \mathbf{\Delta}_{a_1} + 0.4 \cdot \mathbf{\Delta}_{a_2}$	2.790(±0.0330)	2.494(±0.0967)
$0.7 \cdot \mathbf{\Delta}_{a_1} + 0.3 \cdot \mathbf{\Delta}_{a_2}$	3.240(±0.0330)	2.380(±0.0522)
$0.8 \cdot \mathbf{\Delta}_{a_1} + 0.2 \cdot \mathbf{\Delta}_{a_2}$	3.596(±0.0295)	1.940(±0.0432)
$0.9 \cdot \mathbf{\Delta}_{a_1} + 0.1 \cdot \mathbf{\Delta}_{a_2}$	3.510(±0.0630)	1.620(±0.0082)
$1.0 \cdot \mathbf{\Delta}_{a_1} + 0.0 \cdot \mathbf{\Delta}_{a_2}$	3.716(±0.0035)	1.290(±0.0397)

(b) Weighted experience: $\mathbf{\Delta}_{a_1}$: UK, $\mathbf{\Delta}_{a_2}$: Surprised

Table 18: Weighted experiences across different attribute pairs.

Combination	AMOS-a_1 ↑	AMOS-a_2 ↑
$0.0 \cdot \mathbf{\Delta}_{a_1} + 1.0 \cdot \mathbf{\Delta}_{a_2}$	1.270(±0.0345)	4.010(±0.0118)
$0.1 \cdot \mathbf{\Delta}_{a_1} + 0.9 \cdot \mathbf{\Delta}_{a_2}$	1.508(±0.0121)	3.930(±0.0145)
$0.2 \cdot \mathbf{\Delta}_{a_1} + 0.8 \cdot \mathbf{\Delta}_{a_2}$	1.444(±0.0089)	3.762(±0.0290)
$0.3 \cdot \mathbf{\Delta}_{a_1} + 0.7 \cdot \mathbf{\Delta}_{a_2}$	2.048(±0.0161)	3.748(±0.0038)
$0.4 \cdot \mathbf{\Delta}_{a_1} + 0.6 \cdot \mathbf{\Delta}_{a_2}$	2.300(±0.0513)	3.406(±0.0057)
$0.5 \cdot \mathbf{\Delta}_{a_1} + 0.5 \cdot \mathbf{\Delta}_{a_2}$	2.562(±0.0080)	3.220(±0.0158)
$0.6 \cdot \mathbf{\Delta}_{a_1} + 0.4 \cdot \mathbf{\Delta}_{a_2}$	3.176(±0.0339)	2.890(±0.0468)
$0.7 \cdot \mathbf{\Delta}_{a_1} + 0.3 \cdot \mathbf{\Delta}_{a_2}$	3.306(±0.0164)	2.276(±0.0596)
$0.8 \cdot \mathbf{\Delta}_{a_1} + 0.2 \cdot \mathbf{\Delta}_{a_2}$	3.474(±0.0514)	2.640(±0.0430)
$0.9 \cdot \mathbf{\Delta}_{a_1} + 0.1 \cdot \mathbf{\Delta}_{a_2}$	3.520(±0.0133)	1.890(±0.0705)
$1.0 \cdot \mathbf{\Delta}_{a_1} + 0.0 \cdot \mathbf{\Delta}_{a_2}$	3.812(±0.0157)	1.420(±0.0270)

(a) Weighted experience: $\mathbf{\Delta}_{a_1}$: UK, $\mathbf{\Delta}_{a_2}$: >60

Combination	AMOS-a_1 ↑	AMOS-a_2 ↑
$0.0 \cdot \mathbf{\Delta}_{a_1} + 1.0 \cdot \mathbf{\Delta}_{a_2}$	3.190(±0.0205)	3.598(±0.0023)
$0.1 \cdot \mathbf{\Delta}_{a_1} + 0.9 \cdot \mathbf{\Delta}_{a_2}$	3.210(±0.0043)	3.524(±0.0121)
$0.2 \cdot \mathbf{\Delta}_{a_1} + 0.8 \cdot \mathbf{\Delta}_{a_2}$	3.386(±0.0197)	3.280(±0.0520)
$0.3 \cdot \mathbf{\Delta}_{a_1} + 0.7 \cdot \mathbf{\Delta}_{a_2}$	3.426(±0.0088)	3.156(±0.0149)
$0.4 \cdot \mathbf{\Delta}_{a_1} + 0.6 \cdot \mathbf{\Delta}_{a_2}$	3.544(±0.0089)	2.934(±0.0027)
$0.5 \cdot \mathbf{\Delta}_{a_1} + 0.5 \cdot \mathbf{\Delta}_{a_2}$	3.548(±0.0315)	2.730(±0.0320)
$0.6 \cdot \mathbf{\Delta}_{a_1} + 0.4 \cdot \mathbf{\Delta}_{a_2}$	3.638(±0.0032)	2.670(±0.0073)
$0.7 \cdot \mathbf{\Delta}_{a_1} + 0.3 \cdot \mathbf{\Delta}_{a_2}$	3.710(±0.0142)	2.290(±0.0173)
$0.8 \cdot \mathbf{\Delta}_{a_1} + 0.2 \cdot \mathbf{\Delta}_{a_2}$	3.794(±0.0247)	2.434(±0.0121)
$0.9 \cdot \mathbf{\Delta}_{a_1} + 0.1 \cdot \mathbf{\Delta}_{a_2}$	3.810(±0.0193)	1.886(±0.1095)
$1.0 \cdot \mathbf{\Delta}_{a_1} + 0.0 \cdot \mathbf{\Delta}_{a_2}$	4.114(±0.0489)	1.510(±0.0430)

(b) Weighted experience: $\mathbf{\Delta}_{a_1}$: US, $\mathbf{\Delta}_{a_2}$: 20s

Table 19: $\mathbf{\Delta}_{a_1}$: US, $\mathbf{\Delta}_{a_2}$: Teens

Subtraction	AMOS-a_1 ↑	AMOS-a_2 ↑	Win-Attr-a_1 ↑	Win-Attr-a_2 ↑	Win-Qual↑
GT	3.918(±0.0221)	3.540(±0.0193)	-	-	-
Prompt	3.290(±0.0055)	2.790(±0.0143)	0.314(±0.0011)	0.198(±0.0047)	0.364(±0.0082)
$\mathbf{\Delta}_{a_1+a_2}$	3.484(±0.0165)	3.476(±0.0156)	0.336(±0.0016)	0.320(±0.0048)	0.394(±0.0027)
$\mathbf{\Delta}_{a_1}$	3.702(±0.0250)	2.780(±0.0420)	0.410±(0.0012)	0.142(±0.0024)	0.356(±0.0059)
$\mathbf{\Delta}_{a_2}$	2.960(±0.0655)	3.420(±0.0308)	0.274(±0.0042)	0.292(±0.0020)	0.384(±0.0033)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	3.624(±0.0015)	2.608(±0.0222)	0.372(±0.0024)	0.216(±0.0024)	0.294(±0.0023)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	3.090(±0.0780)	3.350(±0.0500)	0.292(±0.0036)	0.338(±0.0021)	0.308(±0.0038)

Table 20: $\mathbf{\Delta}_{a_1}$: US, $\mathbf{\Delta}_{a_2}$: Over 60

Subtraction	AMOS-a_1 ↑	AMOS-a_2 ↑	Win-Attr-a_1 ↑	Win-Attr-a_2 ↑	Win-Qual↑
GT	3.828(±0.0152)	4.020(±0.0258)	-	-	-
Prompt	3.270(±0.0158)	2.596(±0.0265)	0.290(±0.0044)	0.120(±0.0024)	0.372(±0.0044)
$\mathbf{\Delta}_{a_1+a_2}$	3.538(±0.0228)	3.402(±0.0315)	0.296(±0.0029)	0.358(±0.0087)	0.352(±0.0016)
$\mathbf{\Delta}_{a_1}$	3.798(±0.0200)	2.332(±0.0149)	0.426(±0.0017)	0.234(±0.0021)	0.356(±0.0019)
$\mathbf{\Delta}_{a_2}$	2.894(±0.0357)	3.528(±0.0127)	0.220(±0.0033)	0.488(±0.0038)	0.348(±0.0007)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	3.702(±0.0088)	2.240(±0.0518)	0.388(±0.0010)	0.244(±0.0026)	0.330(±0.0017)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	3.270(±0.0395)	3.690(±0.0455)	0.198(±0.0044)	0.372(±0.0051)	0.378(±0.0047)

Table 21: $\mathbf{\Delta}_{a_1}$: 20, $\mathbf{\Delta}_{a_2}$: US

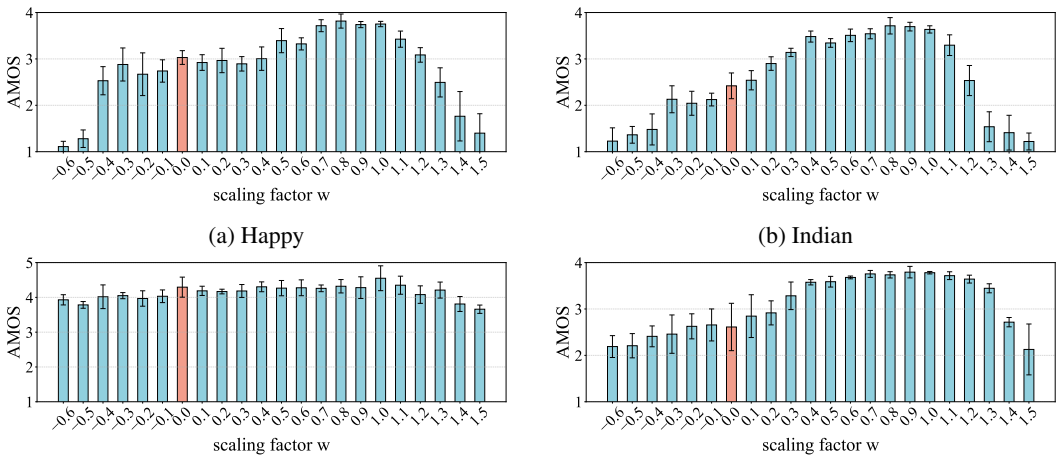
Subtraction	AMOS-a_1 ↑	AMOS-a_2 ↑	Win-Attr-a_1 ↑	Win-Attr-a_2 ↑	Win-Qual↑
GT	3.816(±0.0160)	3.838(±0.0315)	-	-	-
Prompt	2.990(±0.0193)	3.268(±0.0604)	0.206(±0.0048)	0.242(±0.0026)	0.378(±0.0032)
$\mathbf{\Delta}_{a_1+a_2}$	3.614(±0.0092)	3.544(±0.0424)	0.356(±0.0047)	0.354(±0.0039)	0.362(±0.0027)
$\mathbf{\Delta}_{a_1}$	2.558(±0.0138)	3.282(±0.4114)	0.172(±0.0035)	0.378(±0.0024)	0.374(±0.0032)
$\mathbf{\Delta}_{a_2}$	3.454(±0.0115)	3.260(±0.0843)	0.316(±0.0023)	0.316(±0.0019)	0.348(±0.0021)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	2.340(±0.0543)	3.150(±0.5138)	0.196(±0.0024)	0.358(±0.0022)	0.332(±0.0015)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	3.410(±0.0130)	3.320(±0.0220)	0.340(±0.0013)	0.322(±0.0018)	0.346(±0.0020)

Table 22: $\mathbf{\Delta}_{a_1}$: UK, $\mathbf{\Delta}_{a_2}$: 30

Subtraction	AMOS-a_1 ↑	AMOS-a_2 ↑	Win-Attr-a_1 ↑	Win-Attr-a_2 ↑	Win-Qual↑
GT	4.000(±0.0188)	3.854(±0.0060)	-	-	-
Prompt	3.528(±0.0053)	3.610(±0.0109)	0.292(±0.0041)	0.292(±0.0010)	0.444(±0.0051)
$\mathbf{\Delta}_{a_1+a_2}$	3.824(±0.0061)	3.768(±0.0100)	0.376(±0.0030)	0.334(±0.0024)	0.420(±0.0005)
$\mathbf{\Delta}_{a_1}$	3.740(±0.0293)	3.560(±0.0517)	0.342(±0.0058)	0.316(±0.0036)	0.368(±0.0019)
$\mathbf{\Delta}_{a_2}$	1.960(±0.0692)	3.703(±0.0131)	0.152(±0.0019)	0.402(±0.0023)	0.384(±0.0022)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	3.602(±0.0215)	3.556(±0.0319)	0.304(±0.0020)	0.294(±0.0016)	0.384(±0.0018)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	2.166(±0.0603)	3.614(±0.0495)	0.204(±0.0026)	0.380(±0.0029)	0.356(±0.0047)

Table 23: Overall of subtract experience

Subtraction	AMOS-a_1 ↑	AMOS-a_2 ↑	Win-Attr-a_1 ↑	Win-Attr-a_2 ↑	Win-Qual↑
GT	3.891(±0.0180)	3.813(±0.0206)	-	-	-
Prompt	3.270(±0.0114)	3.066(±0.0280)	0.276(±0.0036)	0.213(±0.0026)	0.390(±0.0052)
$\mathbf{\Delta}_{a_1+a_2}$	3.615(±0.0137)	3.548(±0.0249)	0.341(±0.0031)	0.342(±0.0050)	0.382(±0.0019)
$\mathbf{\Delta}_{a_1}$	3.450(±0.0220)	2.989(±0.1300)	0.338(±0.0030)	0.268(±0.0026)	0.364(±0.0032)
$\mathbf{\Delta}_{a_2}$	2.817(±0.0455)	3.478(±0.0352)	0.241(±0.0029)	0.375(±0.0025)	0.366(±0.0021)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_2}$	3.317(±0.0215)	2.889(±0.1549)	0.315(±0.0019)	0.278(±0.0022)	0.335(±0.0018)
$\mathbf{\Delta}_{a_1+a_2} - \mathbf{\Delta}_{a_1}$	2.984(±0.0477)	3.494(±0.0417)	0.259(±0.0030)	0.353(±0.0030)	0.347(±0.0038)



(a) Happy (b) Indian

(c) Male (d) Teens

Figure 5: MOS Results of the Scaling Experiment across different conditions.

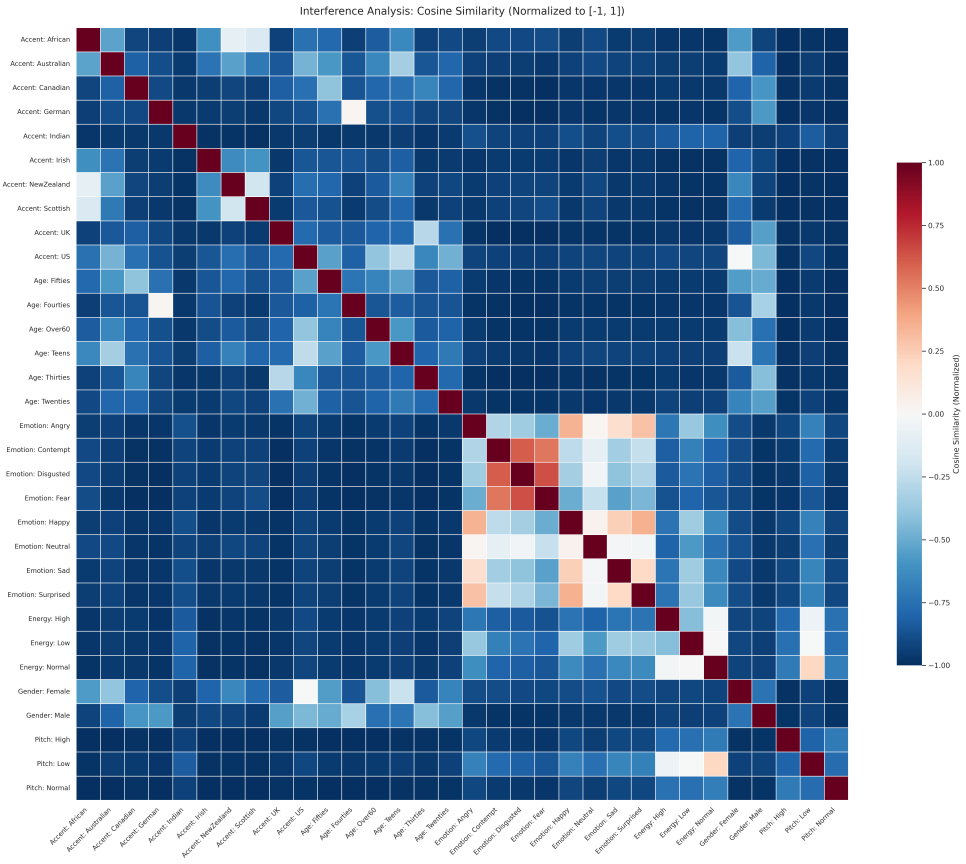


Figure 6: Heatmap of Model Performance Across All Attribute.

H: Cosine Similarity Between all Adapter

I: Visualization of AMOS Scores from Table 3, showing the impact of varying weights between two LoRA modules

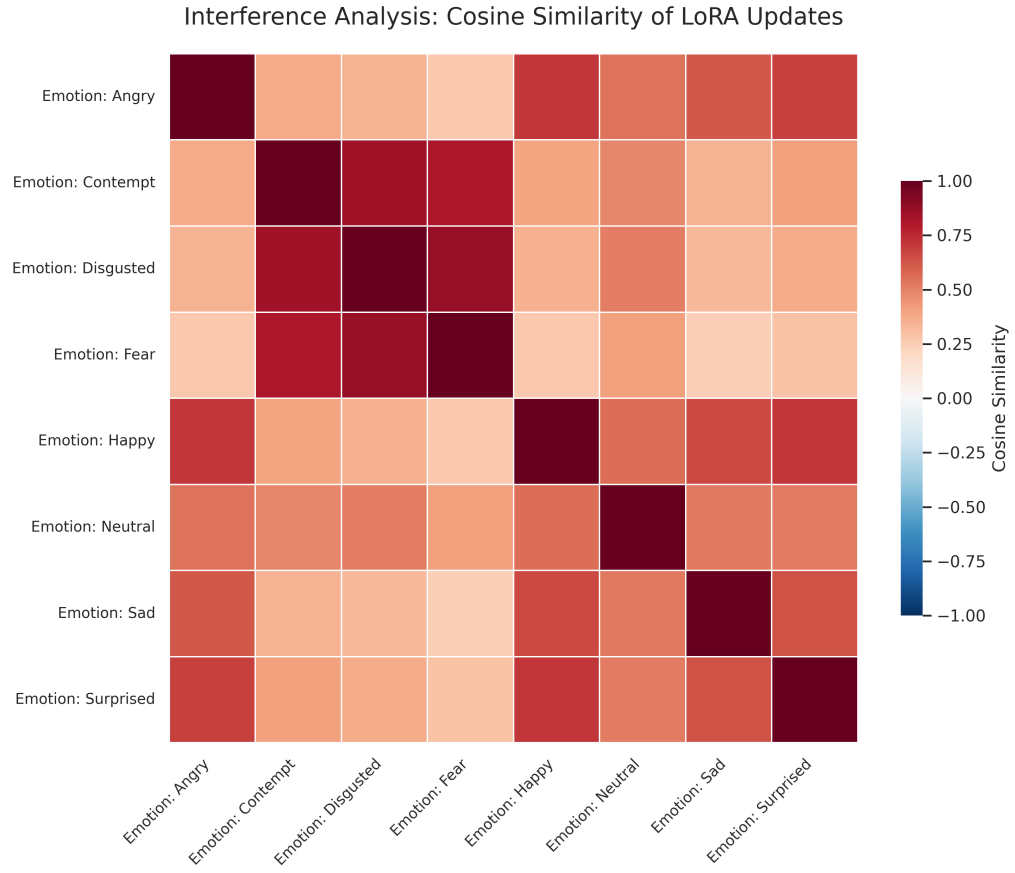


Figure 7: Heatmap visualization of all experimental results.

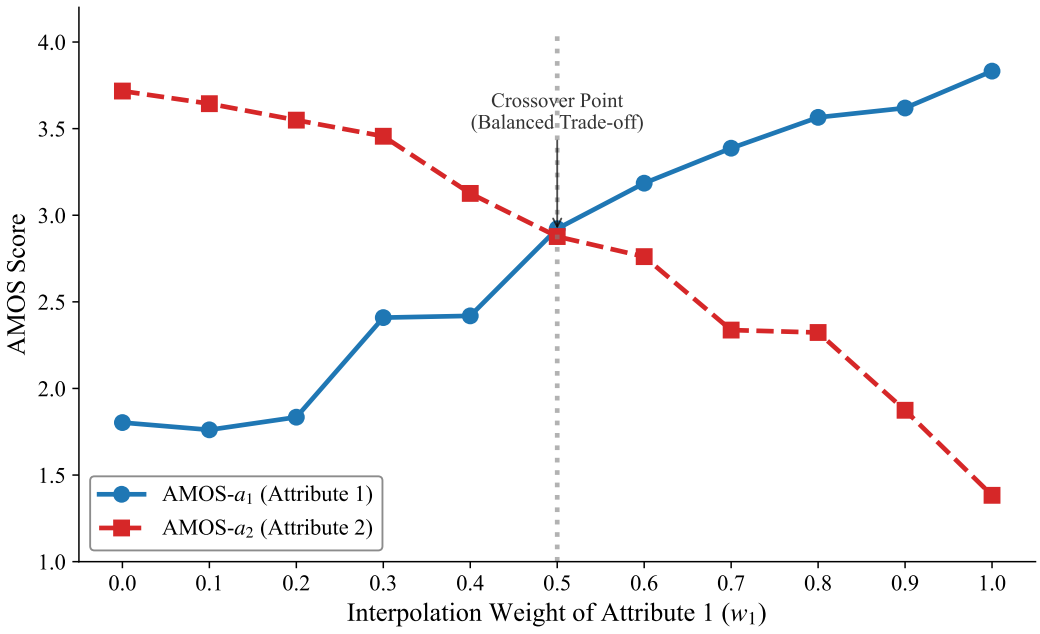
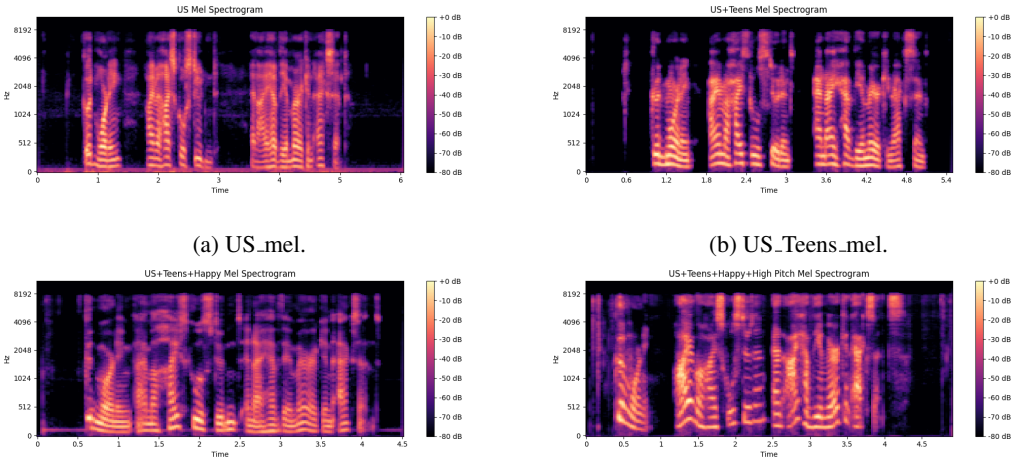


Figure 8: Visualization of AMOS Scores from Table 3, showing the impact of varying weights between two LoRA modules.

J: Failure Modes Analysis



(a) US_mel.

(b) US_Teens_mel.

(c) US_Teens_happy_mel.

(d) US_Teens_happy_high_pitch_mel.

Figure 9: Visualization of Synthesis Failure Modes. The increasing blur and noise in the Mel spectrograms demonstrate the model's difficulty in coordinating multiple attributes, leading to a collapse in output fidelity.

As evidenced by the Mel spectrograms in Figure 9, the model's performance deteriorates with the number of attribute constraints. This decline manifests as a progressive blurring of the spectral structure, where distinct harmonic components begin to smear and unresolved noise becomes more prominent.

K. Statistical Reliability of Evaluation Metrics

To ensure that our subjective metrics (AMOS) and LLM-based proxy metrics (Win-Attr/Win-Qual) are statistically robust and not driven by noise or evaluator bias, we conducted rigorous ground-truth validation studies.

AMOS Inter-Rater Reliability We clarify our evaluation protocol: the reported results are based on a Many-to-One MOS protocol. For every condition, 16 distinct synthesized clips were evaluated by 10 randomly assigned experts from a pool of 30, resulting in 160 independent observations per score. We measured the agreement among these 30 expert annotators using Krippendorff's Alpha (ordinal). As shown in Table 24, the Alpha score of 0.8362 indicates high inter-rater consistency.

Validation of GPT-4o-Audio-Preview To validate GPT-4o-Audio-Preview as an evaluation proxy, we conducted a Ground-Truth Alignment Study. We sampled 100 audio pairs and compared the LLM's decisions against a consensus established by the majority vote of 30 human experts. We measured agreement using Fleiss' Kappa. The scores of 0.8135 (Win-Attr) and 0.8417 (Win-Qual) fall within the range of "substantial agreement," confirming that the LLM serves as a reliable proxy for human perception in this task.

Table 24: Statistical Reliability Analysis of Evaluation Metrics.

Metric	Validation Setup	Statistical Measure	Score
AMOS	100 samples rated by 30 experts (1-5 scale)	Krippendorff's Alpha (α)	0.8362
Win-Attr	100 pairs (Human Majority vs. GPT-4o)	Fleiss' Kappa (κ)	0.8135
Win-Qual	100 pairs (Human Majority vs. GPT-4o)	Fleiss' Kappa (κ)	0.8417

L. Comparison with Reference-Based SOTA Models

We compared TTS-Hub against state-of-the-art reference-based models, CosyVoice 2 and StyleSpeech, to demonstrate the advantages of our modular approach. We evaluated two settings:

- **Matched Reference:** The baseline models are provided with a reference audio that explicitly contains the target attribute.
- **Blank Reference + Prompt:** The baseline models must generate the target attribute based solely on a style tag/prompt, simulating real-world scenarios where an ideal reference is unavailable.

TTS-Hub operates without any reference audio. Table 25 shows that while reference-based models perform well with matched references, their controllability degrades significantly in the Blank Reference setting. TTS-Hub matches the performance of reference-based models without requiring any reference audio, highlighting its superior flexibility.

Table 25: Comparison with SOTA Reference-Based Models. TTS-Hub achieves comparable performance to matched-reference baselines without requiring auxiliary audio inputs.

Model	Condition	WER ↓	AMOS ↑	Win-Attr ↑	Win-Qual ↑
CosyVoice 2	Matched Reference	4.394	4.048	0.652	0.715
CosyVoice 2	Blank Ref + Prompt	4.322	3.314	0.387	0.464
StyleSpeech	Matched Reference	4.713	3.623	0.609	0.651
StyleSpeech	Blank Ref + Prompt	4.604	3.207	0.362	0.389
TTS-Hub (Ours)	**No Reference Needed**	4.622	3.661	0.600	0.409

M. Extended Ablation Studies

Layer-wise LoRA Insertion Analysis

To identify the optimal insertion strategy, we partitioned the Parler-TTS decoder layers into Bottom (0–8), Mid (9–16), and Top (17–24) groups. As shown in Table 26, different attributes favor different components. High-level characteristics like *Age* are best modeled by Bottom layers, while linguistic features like *Accent* benefit most from FFN modification. However, the Full LoRA configuration provides the most robust performance across attributes.

Table 26: Layer-wise Ablation Study on Parler-TTS.

Attribute	Configuration	AMOS ↑	Win-Attr ↑	Win-Qual ↑	MCD ↓	F0-RMSE ↓	SSIM ↑
Age (20s)	Baseline (Full LoRA)	3.390	0.658	0.438	7.389	73.073	0.338
	Layers 0–8 (Bottom)	**3.582**	**0.712**	**0.489**	**7.056**	**68.442**	**0.355**
	Layers 9–16 (Mid)	3.356	0.642	0.425	7.421	74.115	0.334
	Layers 17–24 (Top)	3.112	0.485	0.356	7.895	82.336	0.312
	FFN Only	2.856	0.324	0.245	8.442	89.112	0.298
	Cross-Attention Only	3.375	0.649	0.431	7.405	73.885	0.336
Accent (US)	Baseline (Full LoRA)	3.736	0.664	0.446	7.282	51.677	0.362
	Layers 0–8 (Bottom)	3.412	0.512	0.388	7.654	55.123	0.335
	Layers 9–16 (Mid)	3.455	0.534	0.392	7.598	54.882	0.341
	Layers 17–24 (Top)	3.021	0.356	0.289	8.102	62.441	0.310
	FFN Only	**3.752**	**0.678**	**0.458**	**7.215**	**50.992**	**0.366**
	Cross-Attention Only	3.356	0.498	0.365	7.712	56.012	0.330

Conflict and Interference Analysis

We investigated the geometry of the LoRA parameter space by analyzing conflicting attributes. **Interference:** Table 27 demonstrates that adding contradictory attributes (e.g., High Pitch + Low

Pitch) results in destructive interference rather than a neutral output, confirming that LoRA vectors are not perfectly orthogonal. **Inverse Asymmetry:** Table 28 shows that subtracting an attribute is not equivalent to adding its opposite (i.e., $-\Delta_{\text{High}} \neq +\Delta_{\text{Low}}$), indicating that attributes affect different feature subspaces.

Table 27: Analysis of Conflicting Attributes (Destructive Interference).

Combination Strategy	Target	AMOS ↑	Win-Qual ↑	MCD ↓	F0-RMSE ↓	SSIM ↑
Base (No Adapter)	Neutral	3.428	—	7.091	66.287	0.304
$\Delta_{\text{High Pitch}} + \Delta_{\text{Low Pitch}}$	Neutral	2.811	0.441	8.672	89.314	0.132
$\Delta_{\text{Age 20}} + \Delta_{\text{Age 60}}$	Neutral	2.739	0.512	9.349	80.737	0.155

Table 28: Subtraction vs. Addition of Opposites (Inverse Asymmetry).

Operation	Target	AMOS ↑	Win-Attr ↑	Win-Qual ↑	MCD ↓	F0-RMSE ↓	SSIM ↑
$+\Delta_{\text{Low Pitch}}$	Low Pitch	3.657	0.512	0.338	7.000	45.805	0.222
$-\Delta_{\text{High Pitch}}$	Low Pitch	2.395	0.410	0.375	9.191	78.834	0.121
$+\Delta_{\text{Age 60}}$	Elder	3.273	0.582	0.504	7.126	52.973	0.364
$-\Delta_{\text{Age 20}}$	Elder	1.972	0.363	0.438	8.773	90.194	0.136

N. Efficiency and Scalability Analysis

Runtime and Memory Overhead

We quantified the computational overhead of composing multiple LoRAs. As shown in Table 29, TTS-Hub exhibits negligible latency overhead (1.01x scaling) regardless of the number of adapters, as composition occurs in the weight space before the forward pass. Peak VRAM usage remains stable.

Table 29: Runtime and Memory Overhead on Parler-TTS.

Config	Avg Latency (s)	Scale	Peak VRAM (MB)	Storage (MB)
Base	**7.10**	**1.00x**	**5827.27**	**-**
+1 LoRA	7.16	1.01x	5836.11	+39.57
+3 LoRA	7.14	1.01x	5831.92	+118.70
+6 LoRA	7.15	1.01x	5830.47	+240.76

Fairness and Data Imbalance

We analyzed performance across high-resource and low-resource attributes (Table 30). Performance degrades on low-resource combinations, confirming that fairness limitations stem primarily from data scarcity rather than the compositional mechanism.

O. Advanced Fusion Strategies

To mitigate interference in complex combinations, we explored a **Mini-MoE Router**, a learnable mechanism that predicts optimal mixing weights. The router successfully separates conflicting attributes (e.g., assigning 1.0 to UK and 0.0 to US) and suppresses redundant experts (assigning 0.0 to "Sad" for non-emotional prompts). As shown in Table 31, the Mini-MoE strategy yields measurable gains over naive arithmetic addition, offering a high-quality alternative for scenarios where training overhead is acceptable.

Table 30: Performance comparison between High-Resource and Low-Resource attribute combinations.

Attribute Combination	AMOS ↑	Win-Attr ↑	Win-Qual ↑
High-Res: US Accent + Age 20-30	3.730	0.592	0.322
High-Res: Indian Accent + Age 30-40	3.685	0.553	0.306
Low-Res: African Accent + Age 20-30	3.237	0.359	0.228
Low-Res: German Accent + Teens	3.464	0.461	0.275

Table 31: Comparison of Naive Arithmetic Addition vs. Learned Mini-MoE Fusion.

Fusion Strategy	AMOS ↑	Win-Attr ↑	Win-Qual ↑
Naive Addition (Baseline)	3.650	0.622	0.354
Learned Mini-MoE (Ours)	**3.679**	**0.700**	**0.371**
Improvement	*+0.029*	*+0.078*	*+0.017*

P. Methodological Comparison

Table 32 contextualizes TTS-Hub within the broader landscape of LoRA-based adaptations. Unlike Dynamic Composition approaches (focused on cross-task transfer) or holistic TTS adaptations (which model style as an indivisible unit), TTS-Hub enables atomic decomposition and deterministic arithmetic control.

Table 32: Methodological Comparison of LoRA-based Approaches.

Category	Works	Methodology	Insight for TTS-Hub
Dynamic Composition	LoRAHub, MoLE	Gradient-free search/routing	Motivates deterministic arithmetic for transparency.
LoRA for TTS (Holistic)	EELE, LoRP-TTS	Holistic style modeling	Motivates attribute decomposition for fine control.
TTS-Hub (Ours)	**TTS-Hub**	**Atomic LoRAs + Arithmetic**	**Enables zero-shot, disentangled attribute creation.**